

- 1 (a) The variation with extension x of the tension F in a spring is shown in Fig. 3.1.

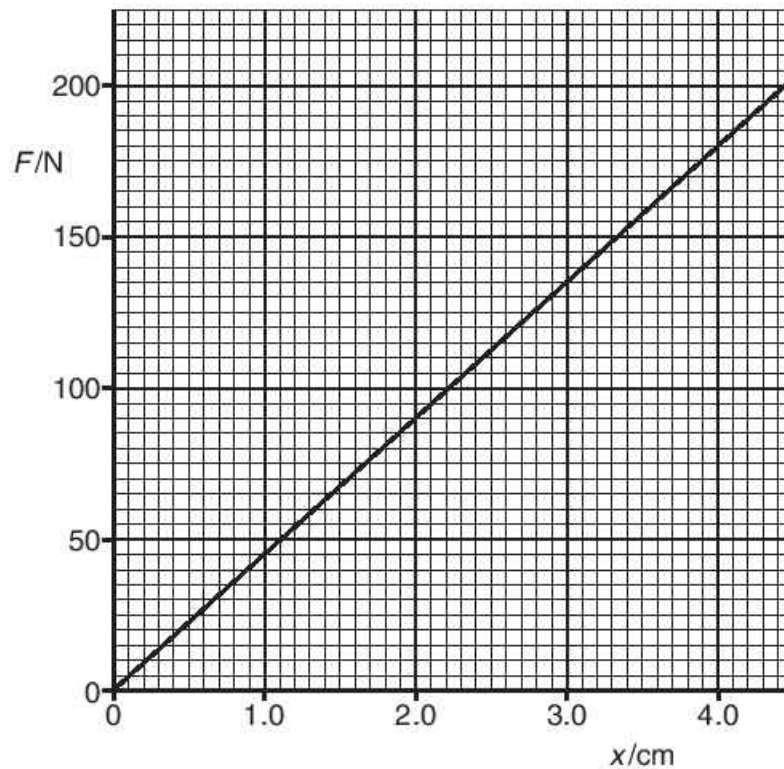


Fig. 3.1

Use Fig. 3.1 to calculate the energy stored in the spring for an extension of 4.0 cm.
Explain your working.

energy = J [3]

- (b) The spring in (a) is used to join together two frictionless trolleys A and B of mass M_1 and M_2 respectively, as shown in Fig. 3.2.

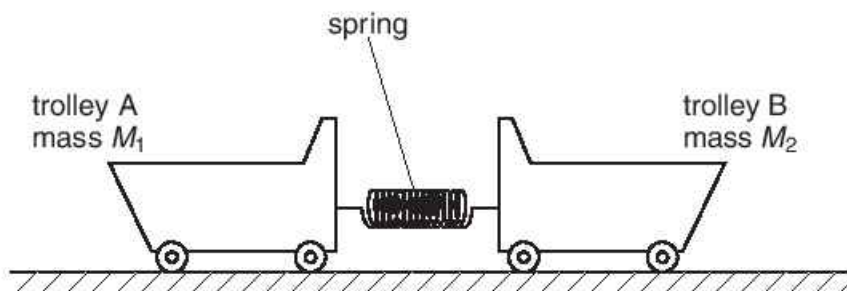


Fig. 3.2

The trolleys rest on a horizontal surface and are held apart so that the spring is extended.

The trolleys are then released.

- (i) Explain why, as the extension of the spring is reduced, the momentum of trolley A is equal in magnitude but opposite in direction to the momentum of trolley B.

.....

.....

.....

..... [2]

- (ii) At the instant when the extension of the spring is zero, trolley A has speed V_1 and trolley B has speed V_2 .
Write down

1. an equation, based on momentum, to relate V_1 and V_2 ,

.....

..... [1]

2. an equation to relate the initial energy E stored in the spring to the final energies of the trolleys.

.....

..... [1]

- (iii) 1. Show that the kinetic energy E_K of an object of mass m is related to its momentum p by the expression

$$E_K = \frac{p^2}{2m}.$$

[1]

2. Trolley A has a larger mass than trolley B.
Use your answer in **(ii) part 1** to deduce which trolley, A or B, has the larger kinetic energy at the instant when the extension of the spring is zero.

.....

..... [1]

- 2 (a) Explain what is meant by *work done*.

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.....
..... [1]

- (b) A car is travelling along a road that has a uniform downhill gradient, as shown in Fig. 2.1.

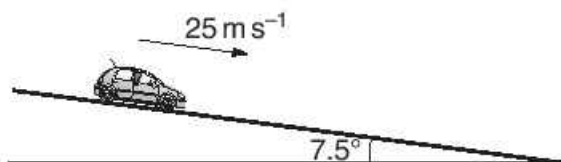


Fig. 2.1

The car has a total mass of 850 kg. The angle of the road to the horizontal is 7.5° . Calculate the component of the weight of the car down the slope.

component of weight = N [2]

- (c) The car in (b) is travelling at a constant speed of 25 m s^{-1} . The driver then applies the brakes to stop the car. The constant force resisting the motion of the car is 4600 N.

- (i) Show that the deceleration of the car with the brakes applied is 4.1 m s^{-2} .

[2]

- (ii) Calculate the distance the car travels from when the brakes are applied until the car comes to rest.

distance = m [2]

- (iii) Calculate

1. the loss of kinetic energy of the car,

loss of kinetic energy = J [2]

2. the work done by the resisting force of 4600 N.

work done = J [1]

- (iv) The quantities in (iii) part 1 and in (iii) part 2 are not equal. Explain why these two quantities are not equal.

.....
..... [1]

3 (a) (i) Explain what is meant by *work done*.

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.....
..... [1]

(ii) Define *power*.

.....
..... [1]

(b) Fig. 3.1 shows part of a fairground ride with a carriage on rails.

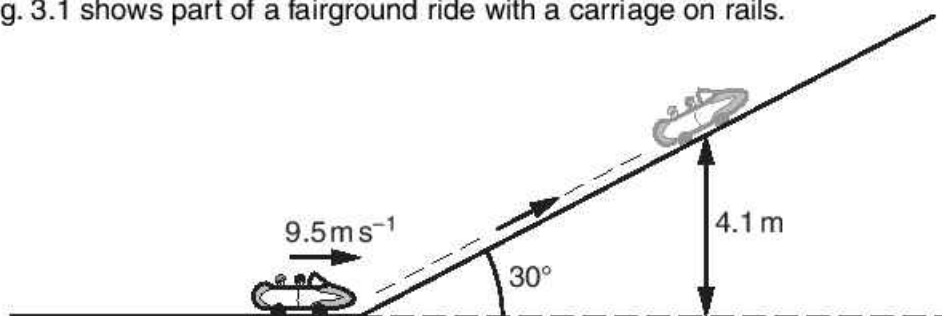


Fig. 3.1

The carriage and passengers have a total mass of 600 kg. The carriage is travelling at a speed of 9.5 m s^{-1} towards a slope inclined at 30° to the horizontal. The carriage comes to rest after travelling up the slope to a vertical height of 4.1 m.

(i) Calculate the kinetic energy, in kJ, of the carriage and passengers as they travel towards the slope.

kinetic energy = kJ [3]

(ii) Show that the gain in potential energy of the carriage and passengers is 24 kJ.

[2]

(iii) Calculate the work done against the resistive force as the carriage moves up the slope.

work done = kJ [1]

(iv) Use your answer in (iii) to calculate the resistive force acting against the carriage as it moves up the slope.

resistive force = N [2]

- 4 (a) Distinguish between *gravitational potential energy* and *electric potential energy*.

.....

 [2]

- (b) A body of mass m moves vertically through a distance h near the Earth's surface. Use the defining equation for work done to derive an expression for the gravitational potential energy change of the body.

[2]

- (c) Water flows down a stream from a reservoir and then causes a water wheel to rotate, as shown in Fig. 4.1.

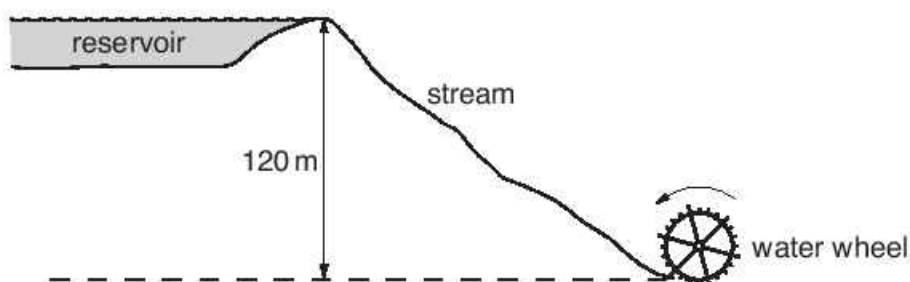


Fig. 4.1

As the water falls through a vertical height of 120 m, gravitational potential energy is converted to different forms of energy, including kinetic energy of the water. At the water wheel, the kinetic energy of the water is only 10% of its gravitational potential energy at the reservoir.

- (i) Show that the speed of the water as it reaches the wheel is 15 ms^{-1} .

[2]

- (ii) The rotating water wheel is used to produce 110 kW of electrical power. Calculate the mass of water flowing per second through the wheel, assuming that the production of electric energy from the kinetic energy of the water is 25% efficient.

mass of water per second = kg s^{-1} [3]

5 (a) Define

9702/22/O/N/11/Q2

(i) *force*,

.....
..... [1]

(ii) *work done*.

.....
..... [1]

(b) A force F acts on a mass m along a straight line for a distance s . The acceleration of the mass is a and the speed changes from an initial speed u to a final speed v .

(i) State the work W done by F .

[1]

(ii) Use your answer in (i) and an equation of motion to show that kinetic energy of a mass can be given by the expression

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2.$$

[3]

(c) A resultant force of 3800 N causes a car of mass of 1500 kg to accelerate from an initial speed of 15 m s^{-1} to a final speed of 30 m s^{-1} .

(i) Calculate the distance moved by the car during this acceleration.

distance = m [2]

(ii) The same force is used to change the speed of the car from 30 m s^{-1} to 45 m s^{-1} . Explain why the distance moved is not the same as that calculated in (i).

.....
.....
..... [1]

- 6 (a) Show that the pressure P due to a liquid of density ρ is proportional to the depth h below the surface of the liquid.

[4]

- (b) The pressure of the air at the top of a mountain is less than that at the foot of the mountain. Explain why the difference in air pressure is not proportional to the difference in height as suggested by the relationship in (a).

.....

.....

.....

..... [2]

7 (a) Define *power*.

9702/23/O/N/12/Q3

.....
..... [1]

(b) A cyclist travels along a horizontal road. The variation with time t of speed v is shown in Fig. 3.1.

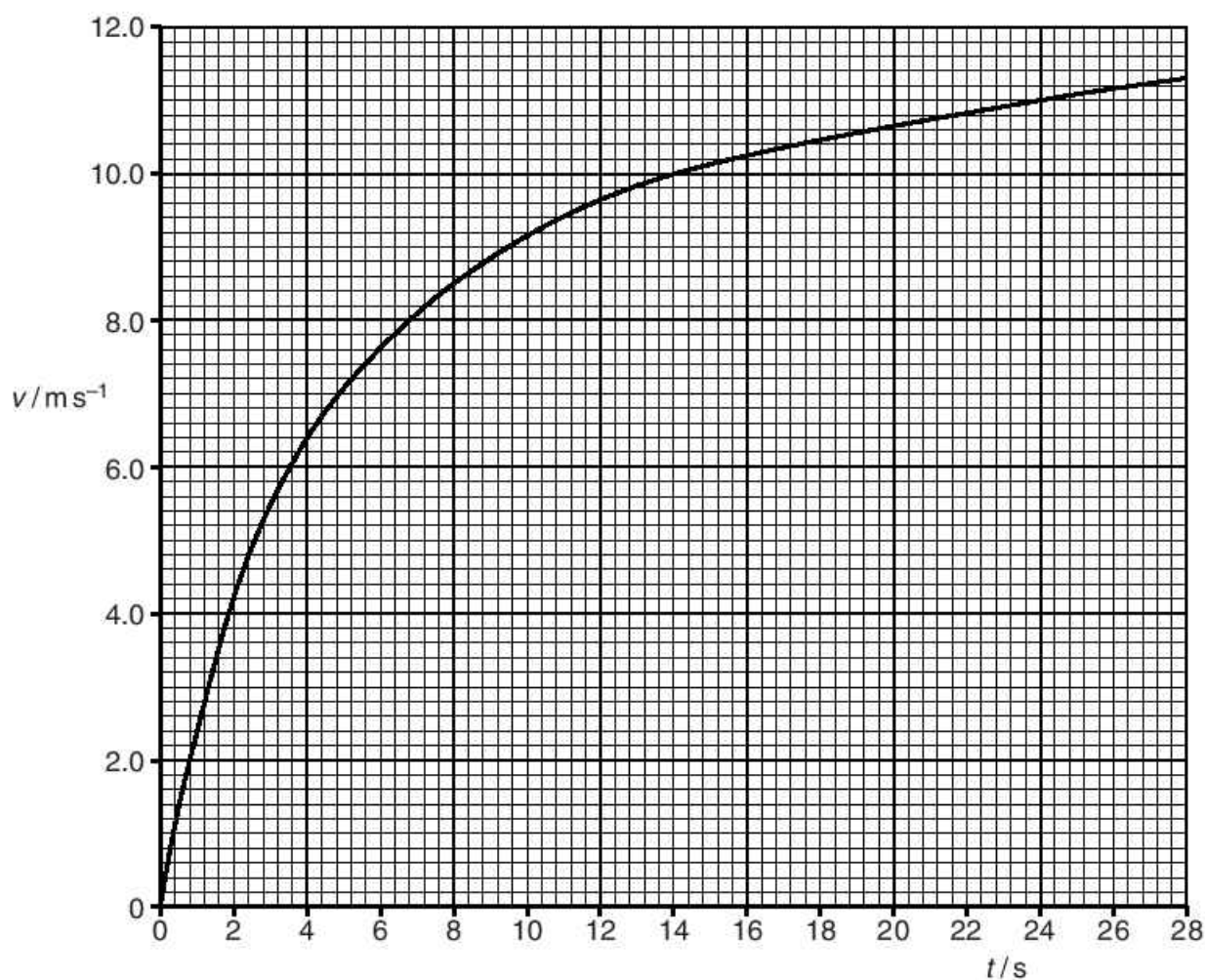


Fig. 3.1

The cyclist maintains a constant power and after some time reaches a constant speed of 12 ms^{-1} .

(i) Describe and explain the motion of the cyclist.

.....
.....
.....
..... [3]

- (ii) When the cyclist is moving at a constant speed of 12 ms^{-1} the resistive force is 48 N . Show that the power of the cyclist is about 600 W . Explain your working.

[2]

- (iii) Use Fig. 3.1 to show that the acceleration of the cyclist when his speed is 8.0 ms^{-1} is about 0.5 ms^{-2} .

[2]

- (iv) The total mass of the cyclist and bicycle is 80 kg . Calculate the resistive force R acting on the cyclist when his speed is 8.0 ms^{-1} . Use the value for the acceleration given in (iii).

$R = \dots\dots\dots \text{ N}$ [3]

- (v) Use the information given in (ii) and your answer to (iv) to show that, in this situation, the resistive force R is proportional to the speed v of the cyclist.

[1]

- 8 (a)** An object falls vertically from rest through air. State and explain the energy conversions that occur as the object falls.

.....

.....

.....

..... [3]

- (b)** A ball of mass 150 g is thrown vertically upwards with an initial speed of 25 ms^{-1} .

- (i)** Calculate the initial kinetic energy of the ball.

kinetic energy = J [3]

- (ii)** The ball reaches a height of 21 m above the point of release.

For the ball rising to this height, calculate

- 1.** the loss of energy of the ball to air resistance,

energy loss = J [3]

- 2.** the average force due to the air resistance.

force = N [2]

- 9 (a) Determine the SI base units of power.

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SI base units of power [3]

- (b) Fig. 1.1 shows a turbine that is used to generate electrical power from the wind.

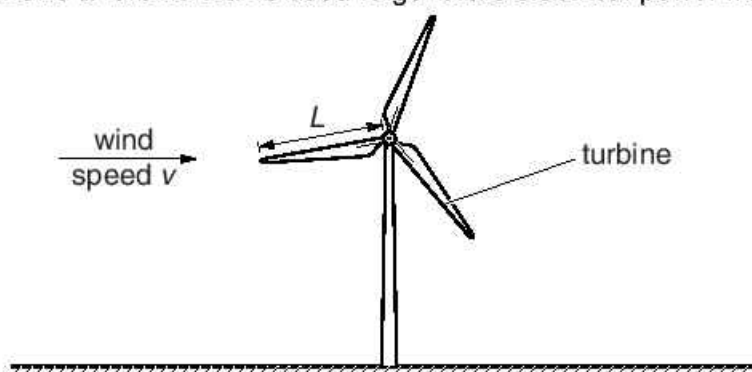


Fig. 1.1

The power P available from the wind is given by

$$P = CL^2\rho v^3$$

where L is the length of each blade of the turbine,
 ρ is the density of air,
 v is the wind speed,
 C is a constant.

- (i) Show that C has no units.

[3]

- (ii) The length L of each blade of the turbine is 25.0 m and the density ρ of air is 1.30 in SI units. The constant C is 0.931.
 The efficiency of the turbine is 55% and the electric power output P is 3.50×10^5 W.
 Calculate the wind speed.

wind speed = ms^{-1} [3]

- (iii) Suggest two reasons why the electrical power output of the turbine is less than the power available from the wind.

1.
2.
-

[2]

10 (a) State what is meant by *work done*.

9702/22/O/N/13/Q3

.....
..... [1]

- (b) A trolley of mass 400 g is moving at a constant velocity of 2.5 m s^{-1} to the right as shown in Fig. 3.1.

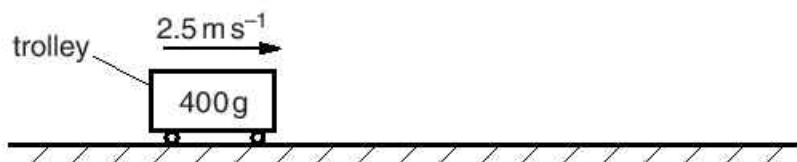


Fig. 3.1

Show that the kinetic energy of the trolley is 1.3 J.

[2]

- (c) The trolley in (b) moves to point P as shown in Fig. 3.2.

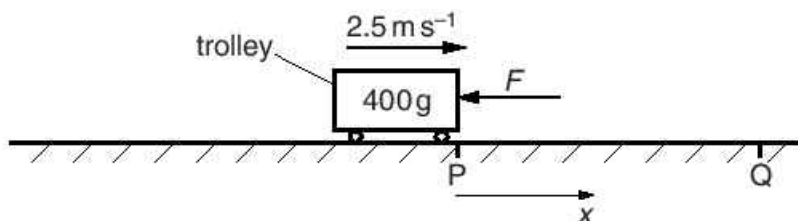


Fig. 3.2

At point P the speed of the trolley is 2.5 m s^{-1} .

A variable force F acts to the left on the trolley as it moves between points P and Q.

The variation of F with displacement x from P is shown in Fig. 3.3.

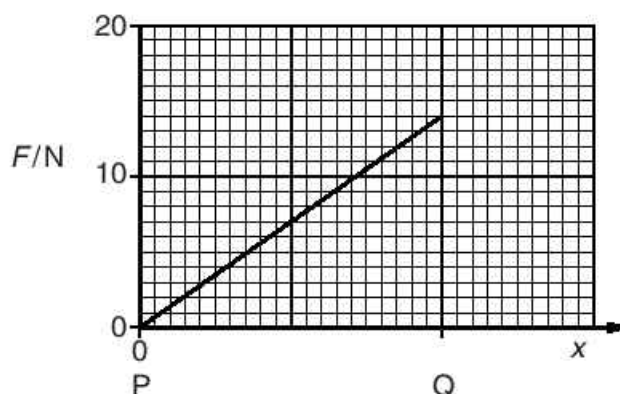


Fig. 3.3

The trolley comes to rest at point Q.

- (i) Calculate the distance PQ.

distance PQ = m [3]

- (ii) On Fig. 3.4, sketch the variation with x of velocity v for the trolley moving between P and Q.

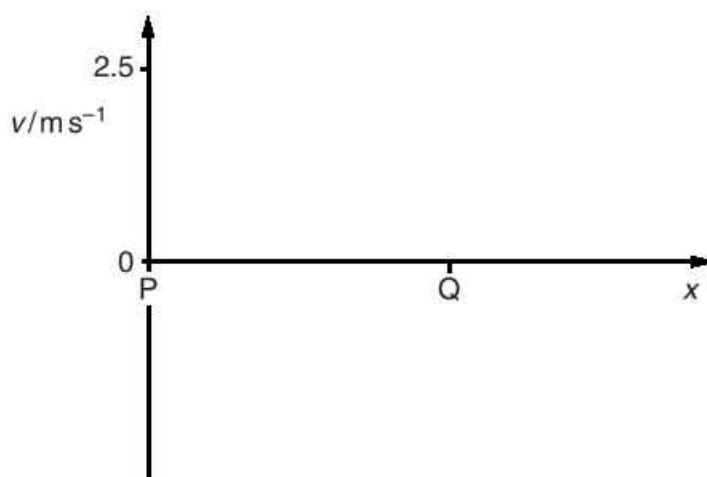


Fig. 3.4

[2]

- 11 (a) Distinguish between gravitational potential energy and elastic potential energy.

.....

[2]

- (b) A ball of mass 65 g is thrown vertically upwards from ground level with a speed of 16 m s^{-1} . Air resistance is negligible.

- (i) Calculate, for the ball,

1. the initial kinetic energy,

kinetic energy = J [2]

2. the maximum height reached.

maximum height = m [2]

- (ii) The ball takes time t to reach maximum height. For time $\frac{t}{2}$ after the ball has been thrown, calculate the ratio

$$\frac{\text{potential energy of ball}}{\text{kinetic energy of ball}}.$$

ratio = [3]

- (iii) State and explain the effect of air resistance on the time taken for the ball to reach maximum height.

.....
[1]

12 (a) (i) Define *power*.

..... [1]

(ii) Use your definition in (i) to show that power may also be expressed as the product of force and velocity.

[2]

(b) A lorry moves up a road that is inclined at 9.0° to the horizontal, as shown in Fig. 2.1.

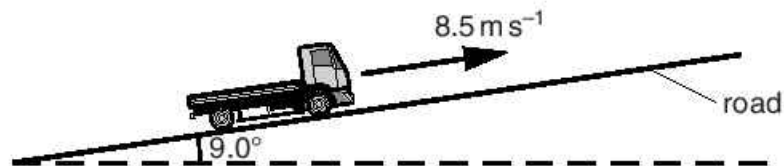


Fig. 2.1

The lorry has mass 2500 kg and is travelling at a constant speed of 8.5 m s^{-1} . The force due to air resistance is negligible.

(i) Calculate the useful power from the engine to move the lorry up the road.

power = kW [3]

(ii) State two reasons why the rate of change of potential energy of the lorry is equal to the power calculated in (i).

1.

.....

2.

.....

[2]

- 13 A metal ball of mass 40 g falls vertically onto a spring, as shown in Fig. 4.1.

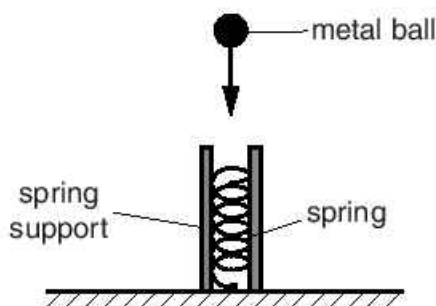


Fig. 4.1 (not to scale)

The spring is supported and stands vertically. The ball has a speed of 2.8 ms^{-1} as it makes contact with the spring. The ball is brought to rest as the spring is compressed.

- (a) Show that the kinetic energy of the ball as it makes contact with the spring is 0.16 J.

[2]

- (b) The variation of the force F acting on the spring with the compression x of the spring is shown in Fig. 4.2.

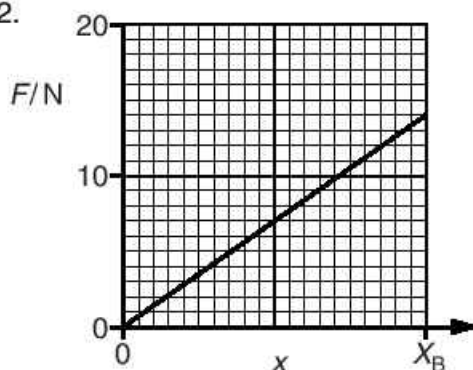


Fig. 4.2

The ball produces a maximum compression X_B when it comes to rest. The spring has a spring constant of 800 N m^{-1} .

Use Fig. 4.2 to

- (i) calculate the compression X_B .

$$X_B = \dots\dots\dots \text{ m [2]}$$

- (ii) show that not all the kinetic energy in (a) is converted into elastic potential energy in the spring.

[2]

- 14 (a) Explain what is meant by *gravitational potential* energy and *kinetic* energy.

gravitational potential energy:

.....

kinetic energy:

.....

[2]

- (b) A ball of mass 400 g is thrown with an initial velocity of 30.0 ms^{-1} at an angle of 45.0° to the horizontal, as shown in Fig. 4.1.

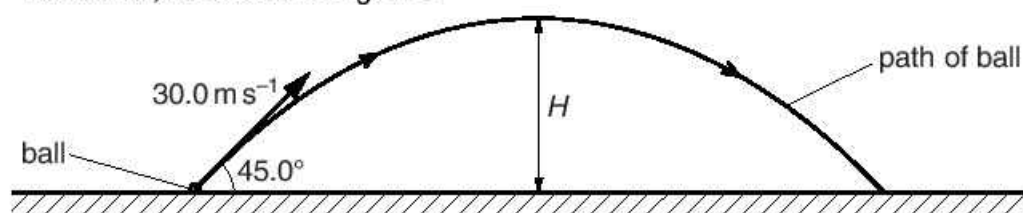


Fig. 4.1

Air resistance is negligible. The ball reaches a maximum height H after a time of 2.16 s.

- (i) Calculate

1. the initial kinetic energy of the ball,

kinetic energy = J [3]

2. the maximum height H of the ball,

$H =$ m [2]

3. the gravitational potential energy of the ball at height H .

potential energy = J [2]

- (ii) 1. Determine the kinetic energy of the ball at its maximum height.

kinetic energy = J [1]

2. Explain why the kinetic energy of the ball at maximum height is not zero.

.....

..... [1]

15 (a) Explain what is meant by *work done*.

9702/23/M/J/14/Q3

.....
.....[1]

(b) A boy on a board B slides down a slope, as shown in Fig. 3.1.

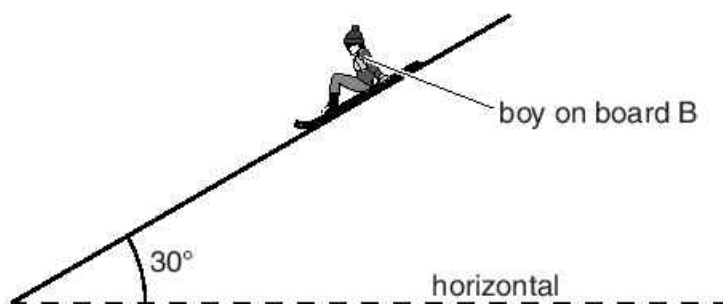


Fig. 3.1

The angle of the slope to the horizontal is 30° . The total resistive force F acting on B is constant.

(i) State a word equation that links the work done by the force F on B to the changes in potential and kinetic energy.

.....
.....[1]

(ii) The boy on the board B moves with velocity v down the slope. The variation with time t of v is shown in Fig. 3.2.

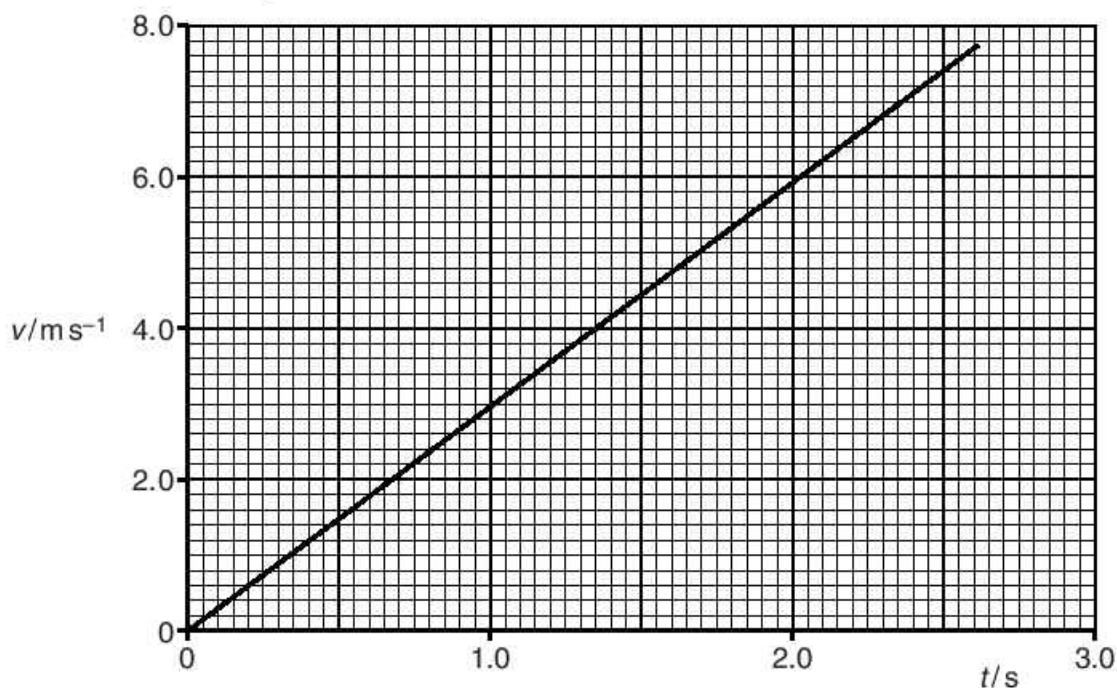


Fig. 3.2

The total mass of B is 75 kg.

For B, from $t = 0$ to $t = 2.5$ s,

1. show that the distance moved down the slope is 9.3 m,

[2]

2. calculate the gain in kinetic energy,

gain in kinetic energy = J [3]

3. calculate the loss in potential energy,

loss in potential energy = J [3]

4. calculate the resistive force F .

$F =$ N [3]

16 A motor is used to move bricks vertically upwards, as shown in Fig. 5.1.

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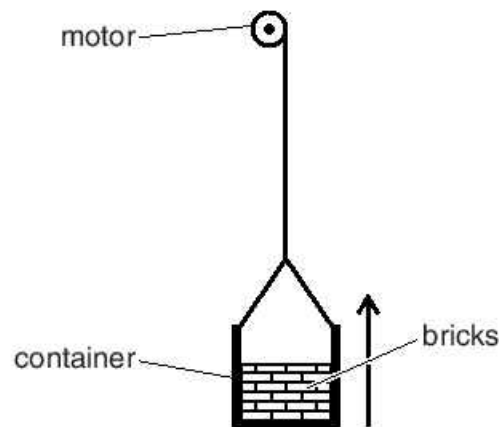


Fig. 5.1

The bricks start from rest and accelerate for 2.0 s. The bricks then travel at a constant speed of 0.64 m s^{-1} for 25 s. Finally the bricks are brought to rest in a further 3.0 s.

The total mass of the bricks is 25 kg.

(a) Determine the change in kinetic energy of the bricks

(i) in the first 2.0 s,

change in kinetic energy = J [2]

(ii) in the next 25 s,

change in kinetic energy = J [1]

(iii) in the final 3.0 s.

change in kinetic energy = J [1]

(b) The bricks are in a container. The weight of the container and bricks is 350 N.
Calculate, for the lifting of the bricks and container when travelling at constant speed,

(i) the gain in potential energy,

energy gain = J [3]

(ii) the power required.

power = W [2]

17 (a) Define *power*.

9702/23/M/J/15/Q3

.....
.....[1]

(b) Fig. 3.1 shows a car travelling at a speed of 22 m s^{-1} on a horizontal road.

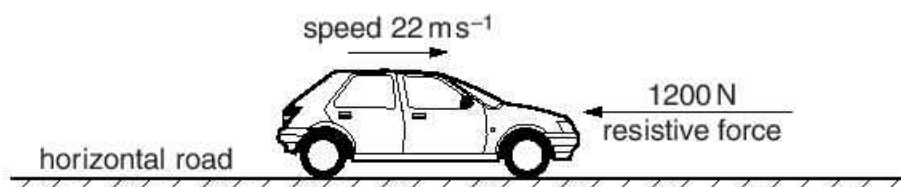


Fig. 3.1

The car has a mass of 1500 kg . A resistive force of 1200 N acts on the car.
Calculate

(i) the force F required from the car to produce an acceleration of 0.82 m s^{-2} ,

$F =$ N [3]

(ii) the power required to produce this acceleration.

power = W [2]

(c) The resistive force on the car is proportional to v^2 , where v is the speed of the car.
Suggest why the car has a maximum speed.

.....
.....[1]

18 (a) State what is meant by

9702/22/F/M/16/Q3

(i) *work done*,

.....
..... [1]

(ii) *elastic potential energy*.

.....
..... [1]

(b) A block of mass 0.40 kg slides in a straight line with a constant speed of 0.30 m s^{-1} along a horizontal surface, as shown in Fig. 3.1.



Fig. 3.1

The block hits a spring and decelerates. The speed of the block becomes zero when the spring is compressed by 8.0 cm .

(i) Calculate the initial kinetic energy of the block.

kinetic energy = J [2]

- (ii) The variation of the compression x of the spring with the force F applied to the spring is shown in Fig. 3.2.

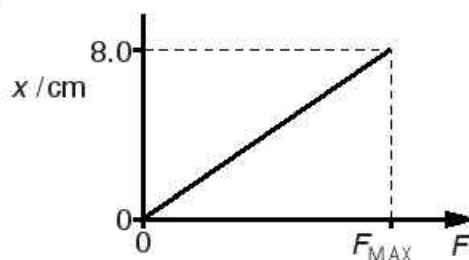


Fig. 3.2

Use your answer in (b)(i) to determine the maximum force F_{MAX} exerted on the spring by the block.
Explain your working.

$$F_{\text{MAX}} = \dots\dots\dots \text{ N [3]}$$

- (iii) Calculate the maximum deceleration of the block.

$$\text{deceleration} = \dots\dots\dots \text{ ms}^{-2} [1]$$

- (iv) State and explain whether the block is in equilibrium

1. before it hits the spring,

.....

2. when its speed becomes zero.

.....

[2]

(c) The energy E stored in a spring is given by

$$E = \frac{1}{2}kx^2$$

where k is the spring constant of the spring and x is its compression.

The mass m of the block in **(b)** is now varied. The initial speed of the block remains constant and the spring continues to obey Hooke's law.

On Fig. 3.3, sketch the variation of the maximum compression x_0 of the spring with mass m .

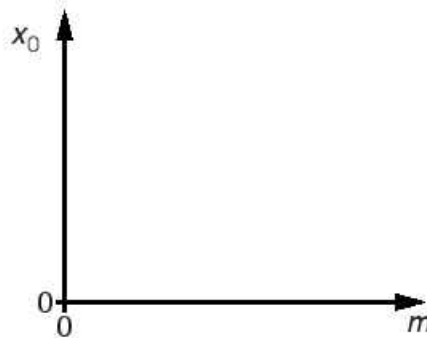


Fig. 3.3

[2]

19 (a) State what is meant by *elastic potential energy*.

9702/21/M/J/16/Q4

.....
.....[1]

(b) A spring is extended by applying a force. The variation with extension x of the force F is shown in Fig. 4.1 for the range of values of x from 20 cm to 40 cm.

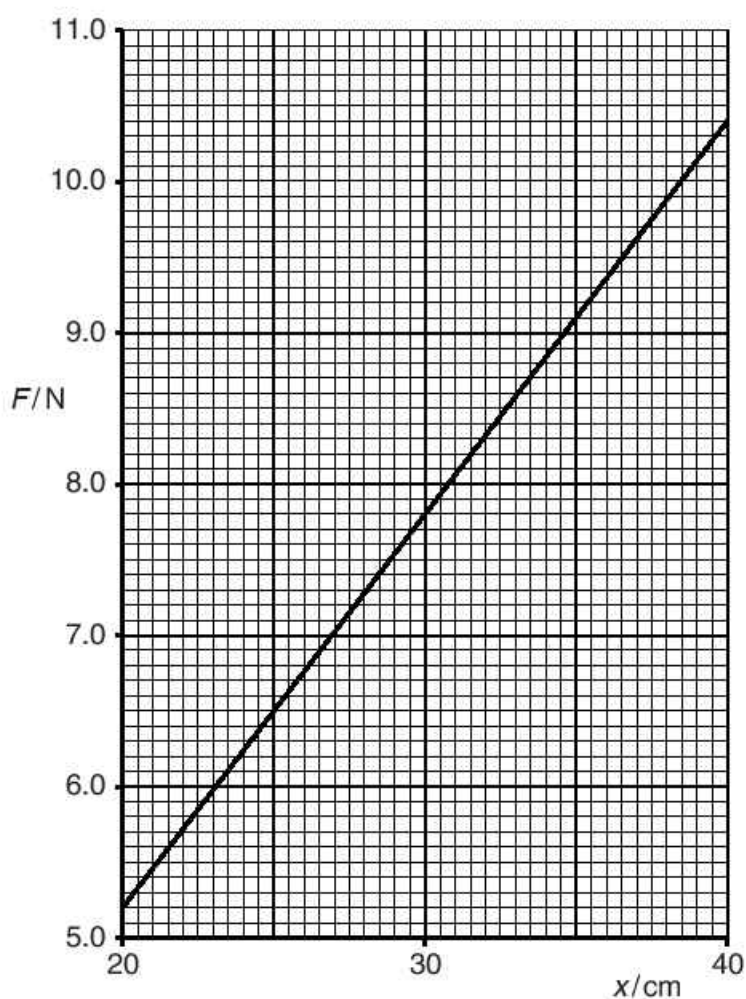


Fig. 4.1

(i) Use data from Fig. 4.1 to show that the spring obeys Hooke's law for this range of extensions.

.....
.....
.....[2]

(ii) Use Fig. 4.1 to calculate

1. the spring constant,

spring constant = Nm^{-1} [2]

2. the work done extending the spring from $x = 20\text{cm}$ to $x = 40\text{cm}$.

work done = J [3]

(c) A force is applied to the spring in **(b)** to give an extension of 50 cm.

State how you would check that the spring has not exceeded its elastic limit.

.....
.....[1]

20 (a) Define *acceleration*.

.....
[1]

(b) A man travels on a toboggan down a slope covered with snow from point A to point B and then to point C. The path is illustrated in Fig. 1.1.

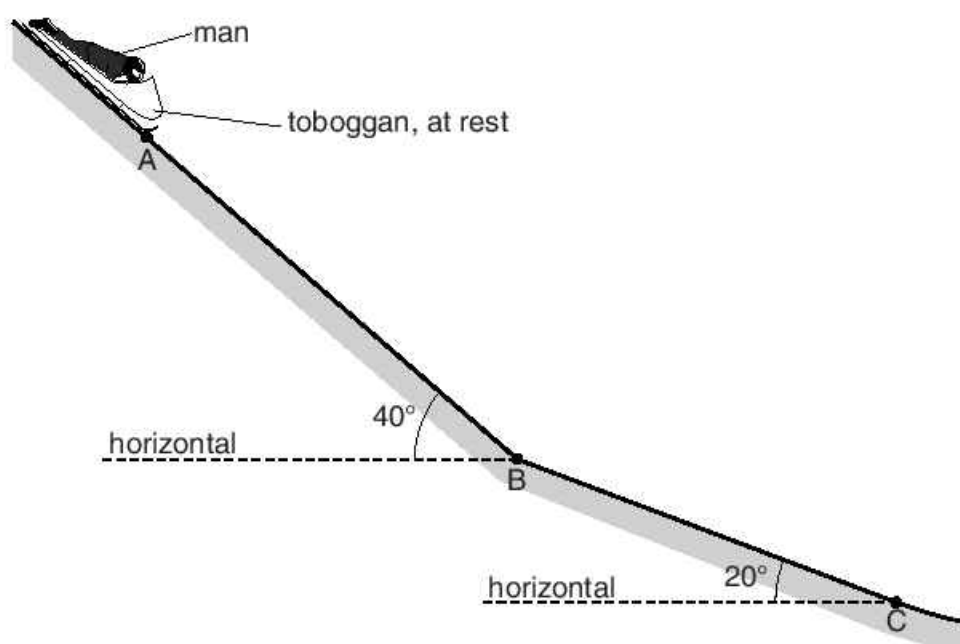


Fig. 1.1 (not to scale)

The slope AB makes an angle of 40° with the horizontal and the slope BC makes an angle of 20° with the horizontal. Friction is not negligible.

The man and toboggan have a combined mass of 95 kg.

The man starts from rest at A and has constant acceleration between A and B. The man takes 19 s to reach B. His speed is 36 ms^{-1} at B.

(i) Calculate the acceleration from A to B.

acceleration = ms^{-2} [2]

(ii) Show that the distance moved from A to B is 340 m.

[1]

(iii) For the man and toboggan moving from A to B, calculate

1. the change in kinetic energy,

change in kinetic energy = J [2]

2. the change in potential energy.

change in potential energy = J [2]

(iv) Use your answers in **(iii)** to determine the average frictional force that acts on the toboggan between A and B.

frictional force = N [2]

(v) A parachute opens on the toboggan as it passes point B. There is a constant deceleration of 3.0 ms^{-2} from B to C.

Calculate the frictional force that produces this deceleration between B and C.

frictional force = N [2]

- 21 (a) Explain what is meant by *gravitational potential energy* and by *kinetic energy*.

gravitational potential energy:

.....

kinetic energy:

.....

[2]

- (b) A motion sensor is used to measure the velocity of a ball falling vertically towards the ground, as illustrated in Fig. 3.1.

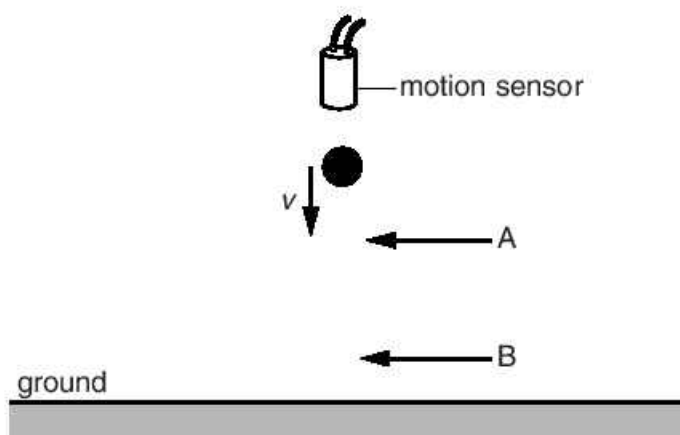


Fig. 3.1

The ball passes through points A and B as it falls. The ball has a mass of 1.5 kg.

The variation with time t of the velocity v of the ball as it falls from A to B is shown in Fig. 3.2.

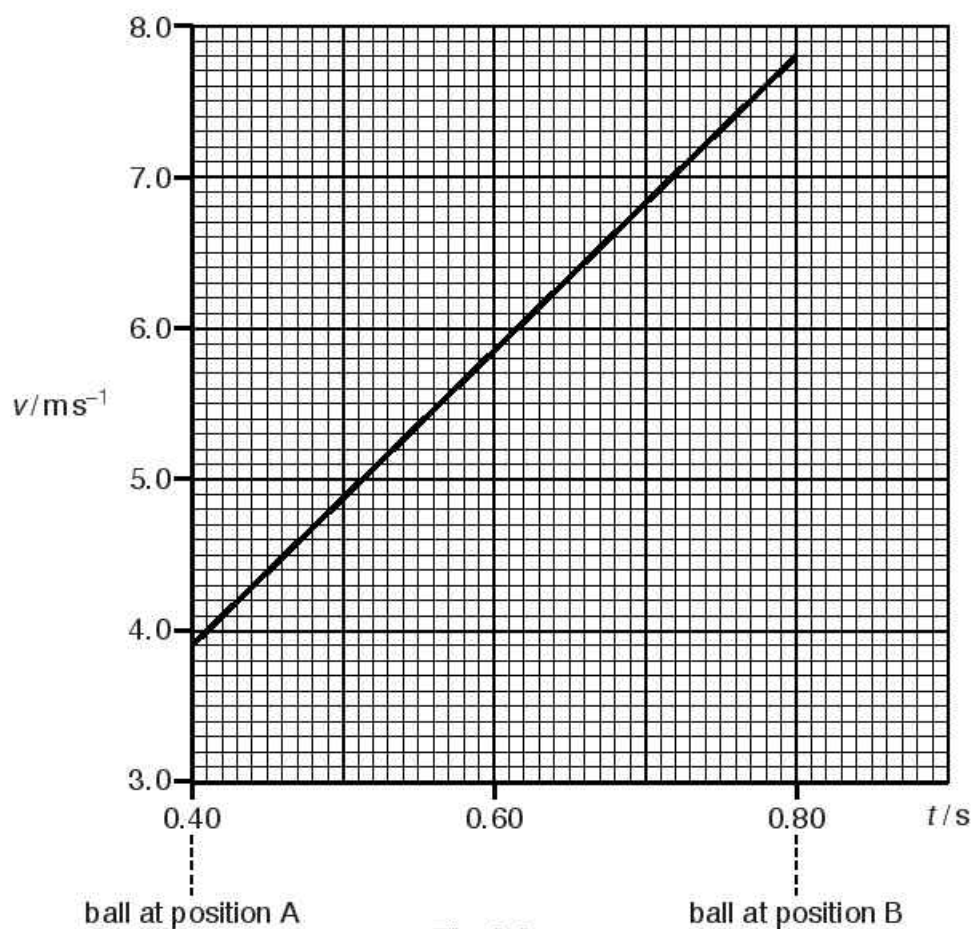


Fig. 3.2

Use Fig. 3.2 to calculate, for the ball falling from A to B,

(i) the displacement,

displacement = m [3]

(ii) the acceleration,

acceleration = ms^{-2} [2]

(iii) the change in kinetic energy.

change in kinetic energy =J [3]

(c) Show that the work done by the gravitational field on the ball in **(b)** as it moves from A to B is equal to the change in kinetic energy.

[2]

22 A ball of mass 0.030 kg moves along a curved track, as shown in Fig. 2.1.

9702/22/O/N/16/Q2

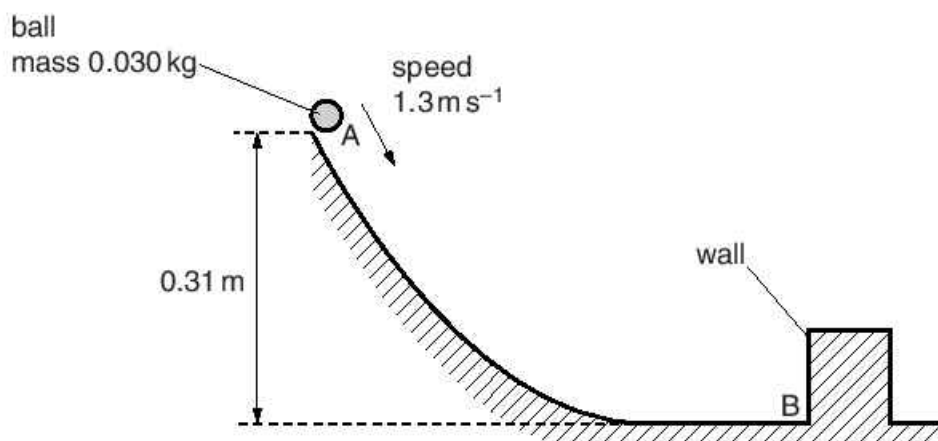


Fig. 2.1

The speed of the ball is 1.3 m s^{-1} when it is at point A at a height of 0.31 m . The ball moves down the track and collides with a vertical wall at point B. The ball then rebounds back up the track. It may be assumed that frictional forces are negligible.

- (a)** Calculate the change in gravitational potential energy of the ball in moving from point A to point B.

change in gravitational potential energy = J [2]

- (b)** Show that the ball hits the wall at B with a speed of 2.8 m s^{-1} .

- (c)** The change in momentum of the ball due to the collision with the wall is $0.096 \text{ kg m s}^{-1}$. The ball is in contact with the wall for a time of 20 ms.

Determine, for the ball colliding with the wall,

- (i)** the speed immediately after the collision,

speed = ms^{-1} [2]

- (ii)** the magnitude of the average force on the ball.

force = N [2]

- (d)** State and explain whether the collision is elastic or inelastic.

.....
.....[1]

- (e)** In practice, frictional effects are significant so that the actual increase in kinetic energy of the ball in moving from A to B is 76 mJ. The length of the track between A and B is 0.60 m.

Use your answer in **(a)** to determine the average frictional force acting on the ball as it moves from A to B.

frictional force = N [2]

- 23 A ball is thrown vertically upwards towards a ceiling and then rebounds, as illustrated in Fig. 3.1.

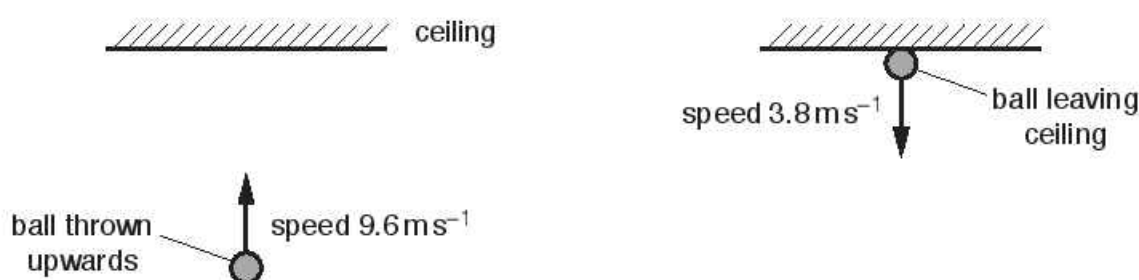


Fig. 3.1

The ball is thrown with speed 9.6 m s^{-1} and takes a time of 0.37 s to reach the ceiling. The ball is then in contact with the ceiling for a further time of 0.085 s until leaving it with a speed of 3.8 m s^{-1} . The mass of the ball is 0.056 kg . Assume that air resistance is negligible.

- (a) Show that the ball reaches the ceiling with a speed of 6.0 m s^{-1} .

[1]

- (b) Calculate the height of the ceiling above the point from which the ball was thrown.

height = m [2]

- (c) Calculate

- (i) the increase in gravitational potential energy of the ball for its movement from its initial position to the ceiling,

increase in gravitational potential energy = J [2]

- (ii) the decrease in kinetic energy of the ball while it is in contact with the ceiling.

decrease in kinetic energy = J [2]

- (d) State how Newton's third law applies to the collision between the ball and the ceiling.

.....
.....
.....
..... [2]

- (e) Calculate the change in momentum of the ball during the collision.

change in momentum = N s [2]

- (f) Determine the magnitude of the average force exerted by the ceiling on the ball during the collision.

average force = N [2]

24 (a) Define

(i) *displacement*,

.....
 [1]

(ii) *acceleration*.

.....
 [1]

(b) A remote-controlled toy car moves up a ramp and travels across a gap to land on another ramp, as illustrated in Fig. 1.1.

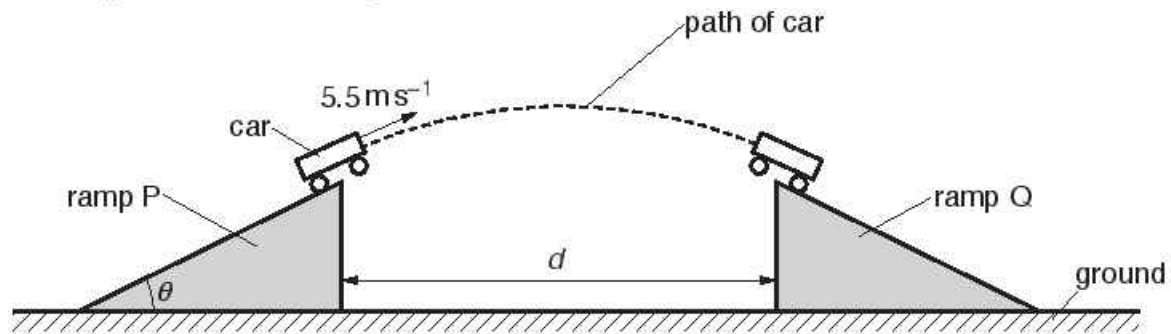


Fig. 1.1

The car leaves ramp P with a velocity of 5.5 ms^{-1} at an angle θ to the horizontal. The horizontal component of the car's velocity as it leaves the ramp is 4.6 ms^{-1} . The car lands at the top of ramp Q. The tops of both ramps are at the same height and are distance d apart. Air resistance is negligible.

(i) Show that the car leaves ramp P with a vertical component of velocity of 3.0 ms^{-1} .

[1]

(ii) Determine the time taken for the car to travel between the ramps.

time taken = s [2]

- (iii) Calculate the horizontal distance d between the tops of the ramps.

$d = \dots\dots\dots$ m [1]

- (iv) Calculate the ratio

$$\frac{\text{kinetic energy of the car at its maximum height}}{\text{kinetic energy of the car as it leaves ramp P}}.$$

ratio = $\dots\dots\dots$ [3]

- (c) Ramp Q is removed. The car again leaves ramp P as in (b) and now lands directly on the ground. The car leaves ramp P at time $t = 0$ and lands on the ground at time $t = T$.

On Fig. 1.2, sketch the variation with time t of the vertical component v_y of the car's velocity from $t = 0$ to $t = T$. Numerical values of v_y and t are not required.

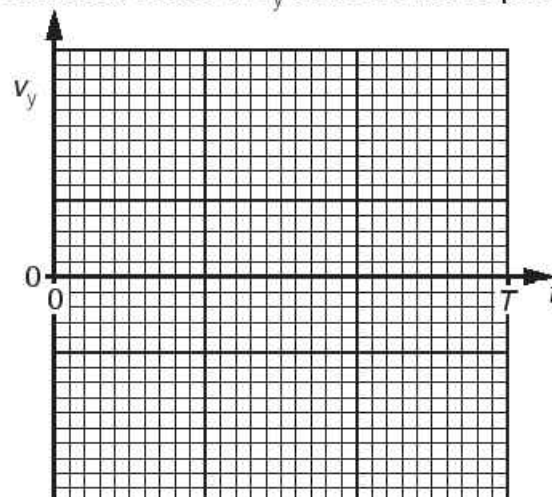


Fig. 1.2

[2]

25 (a) (i) Define *power*.

.....
 [1]

(ii) State what is meant by *gravitational potential energy*.

.....
 [1]

(b) An aircraft of mass 1200 kg climbs upwards with a constant velocity of 45 ms^{-1} , as shown in Fig. 3.1.

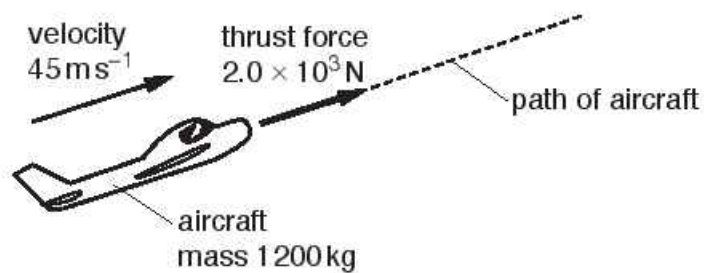


Fig. 3.1 (not to scale)

The aircraft's engine produces a thrust force of $2.0 \times 10^3\text{ N}$ to move the aircraft through the air. The rate of increase in height of the aircraft is 3.3 ms^{-1} .

(i) Calculate the power produced by the thrust force.

power = W [2]

- (ii) Determine, for a time interval of 3.0 minutes,
1. the work done by the thrust force to move the aircraft,

work done = J [2]

2. the increase in gravitational potential energy of the aircraft,

increase in gravitational potential energy = J [2]

3. the work done against air resistance.

work done = J [1]

- (iii) Use your answer in (b)(ii) part 3 to calculate the force due to air resistance acting on the aircraft.

force = N [1]

- (iv) With reference to the motion of the aircraft, state and explain whether the aircraft is in equilibrium.

.....

.....

..... [2]

- 26 A golfer strikes a ball so that it leaves horizontal ground with a velocity of 6.0 m s^{-1} at an angle θ to the horizontal, as illustrated in Fig. 1.1.

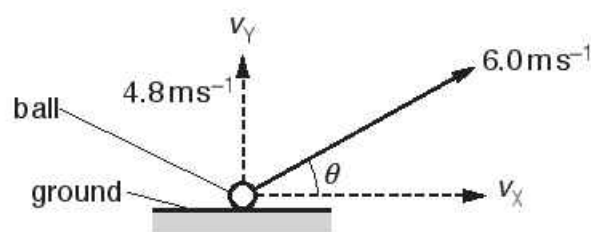


Fig. 1.1 (not to scale)

The magnitude of the initial vertical component v_y of the velocity is 4.8 m s^{-1} . Assume that air resistance is negligible.

- (a) Show that the magnitude of the initial horizontal component v_x of the velocity is 3.6 m s^{-1} .

- (b) The ball leaves the ground at time $t = 0$ and reaches its maximum height at $t = 0.49\text{ s}$.

On Fig. 1.2, sketch separate lines to show the variation with time t , until the ball returns to the ground, of

(i) the vertical component v_y of the velocity (label this line Y), [2]

(ii) the horizontal component v_x of the velocity (label this line X). [2]

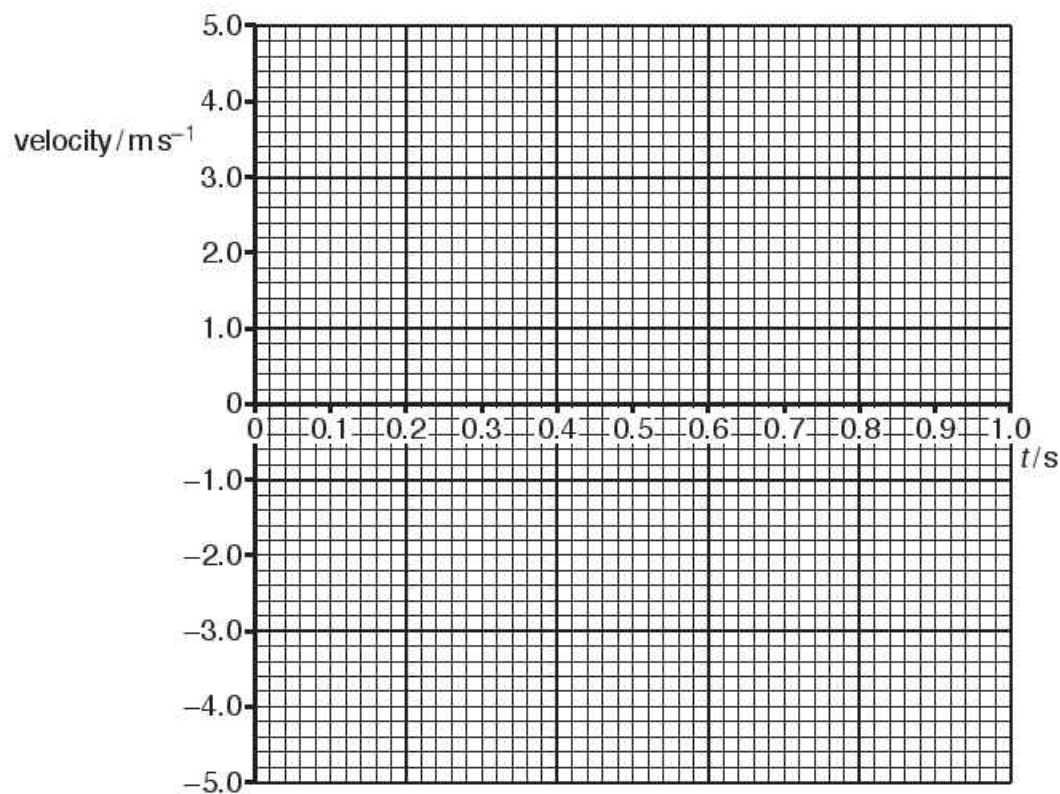


Fig. 1.2

- (c) Calculate the maximum height reached by the ball.

maximum height = m [2]

- (d) For the movement of the ball from the ground to its maximum height, determine the ratio
- $$\frac{\text{kinetic energy at maximum height}}{\text{change in gravitational potential energy}}.$$

ratio = [4]

- (e) In practice, significant air resistance acts on the ball. Explain why the actual time taken for the ball to reach maximum height is less than the time calculated when air resistance is assumed to be negligible.

.....
.....
..... [1]

- 27 (a) State what is meant by *kinetic energy*.

.....

..... [1]

- (b) A cannon fires a shell vertically upwards. The shell leaves the cannon with a speed of 80 m s^{-1} and a kinetic energy of 480 J . The shell then rises to a maximum height of 210 m . The effect of air resistance is significant.

- (i) Show that the mass of the shell is 0.15 kg .

[2]

- (ii) For the movement of the shell from the cannon to its maximum height, calculate

1. the gain in gravitational potential energy,

gain in gravitational potential energy = J [2]

2. the work done against air resistance.

work done = J [1]

- (iii) Determine the average force due to the air resistance acting on the shell as it moves from the cannon to its maximum height.

force = N [2]

- (iv) The shell leaves the cannon at time $t = 0$ and reaches maximum height at time $t = T$.

On Fig. 2.1, sketch the variation with time t of the velocity v of the shell from time $t = 0$ to time $t = T$. Numerical values of v and t are not required.

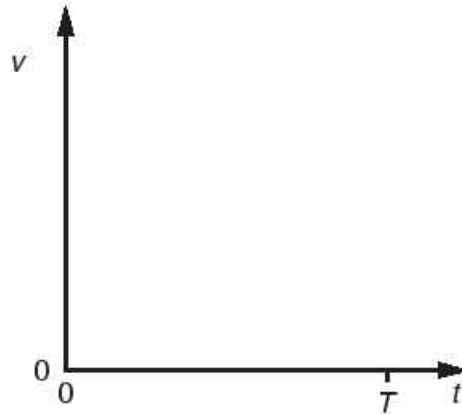


Fig. 2.1

[2]

- (v) The force due to the air resistance is a vector quantity.

Compare the force due to the air resistance acting on the shell as it rises with the force due to the air resistance as it falls.

.....

.....

.....

.....

..... [2]

- 28 The variation with extension x of the force F acting on a spring is shown in Fig. 3.1.

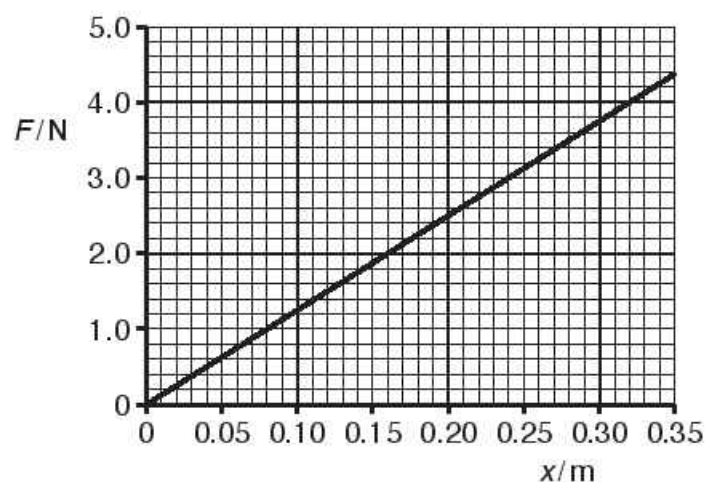


Fig. 3.1

The spring of unstretched length 0.40 m has one end attached to a fixed point, as shown in Fig. 3.2.

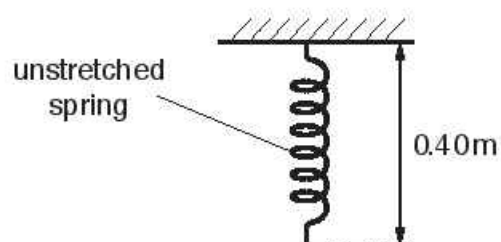


Fig. 3.2

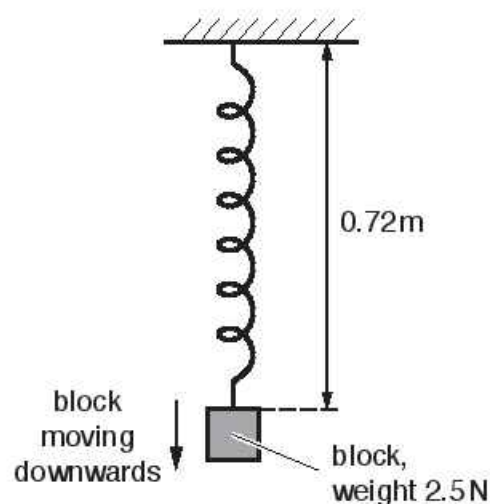


Fig. 3.3

A block of weight 2.5 N is then attached to the spring. The block is then released and begins to move downwards. At one instant, as the block is continuing to move downwards, the spring has a length of 0.72 m, as shown in Fig. 3.3.

Assume that the air resistance and the mass of the spring are both negligible.

(a) For the change in length of the spring from 0.40 m to 0.72 m:

(i) use Fig. 3.1 to show that the increase in elastic potential energy of the spring is 0.64 J

[2]

(ii) calculate the decrease in gravitational potential energy of the block of weight 2.5 N.

decrease in potential energy = J [2]

(b) Use the information in **(a)(i)** and your answer in **(a)(ii)** to determine, for the instant when the length of the spring is 0.72 m:

(i) the kinetic energy of the block

kinetic energy = J [1]

(ii) the speed of the block.

speed = ms^{-1} [2]

- 29 (a) State what is meant by the *centre of gravity* of a body.

.....
 [1]

- (b) A uniform square sign with sides of length 0.68 m is fixed at its corner points A and B to a wall. The sign is also supported by a wire CD, as shown in Fig. 3.1.

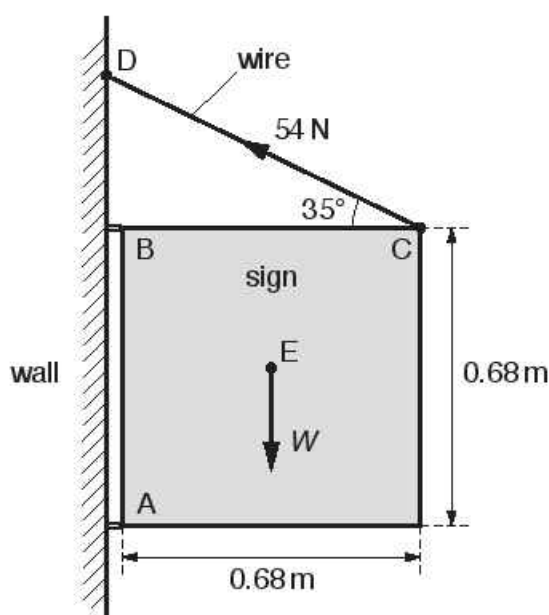


Fig. 3.1 (not to scale)

The sign has weight W and centre of gravity at point E. The sign is held in a vertical plane with side BC horizontal. The wire is at an angle of 35° to side BC. The tension in the wire is 54 N .

The force exerted on the sign at B is only in the vertical direction.

- (i) Calculate the vertical component of the tension in the wire.

vertical component of tension = N [1]

- (ii) Explain why the force on the sign at B does not have a moment about point A.

.....
 [1]

(iii) By taking moments about point A, show that the weight W of the sign is 150 N.

[2]

(iv) Calculate the total vertical force exerted by the wall on the sign at points A and B.

total vertical force = N [1]

(c) The sign in (b) is held together by nuts and bolts. One of the nuts falls vertically from rest through a distance of 4.8 m to the pavement below. The nut lands on the pavement with a speed of 9.2 m s^{-1} .

Determine, for the nut falling from the sign to the pavement, the ratio

$$\frac{\text{change in gravitational potential energy}}{\text{final kinetic energy}}.$$

ratio = [4]

- 30 (a) A resultant force F moves an object of mass m through distance s in a straight line. The force gives the object an acceleration a so that its speed changes from initial speed u to final speed v .

(i) State an expression for:

1. the work W done by the force, in terms of a , m and s

$$W = \dots\dots\dots [1]$$

2. the distance s , in terms of a , u and v .

$$s = \dots\dots\dots [1]$$

(ii) Use your answers in (i) to show that the kinetic energy of the object is given by

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2.$$

Explain your working.

[2]

- (b) A ball of mass 0.040 kg is projected into the air from horizontal ground, as illustrated in Fig. 2.1.

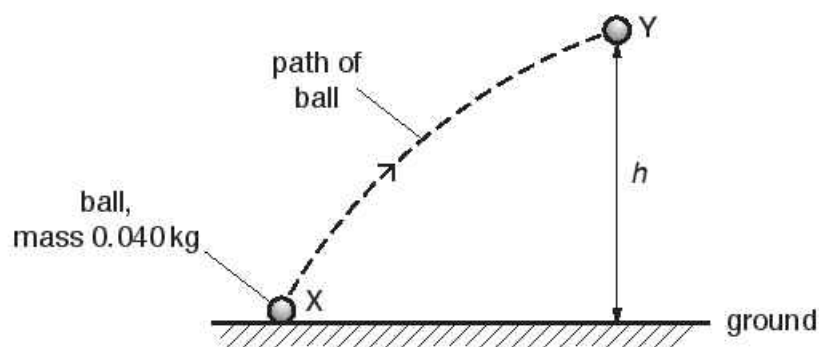


Fig. 2.1

The ball is launched from a point X with a kinetic energy of 4.5 J . At point Y, the ball has a speed of 9.5 m s^{-1} . Air resistance is negligible.

(i) For the movement of the ball from X to Y, draw a solid line on Fig. 2.1 to show:

1. the distance moved (label this line D)
2. the displacement (label this line S).

[2]

(ii) By consideration of energy transfer, determine the height h of point Y above the ground.

$h = \dots\dots\dots$ m [3]

(iii) On Fig. 2.2, sketch the variation of the kinetic energy of the ball with its vertical height above the ground for the movement of the ball from X to Y. Numerical values are not required.

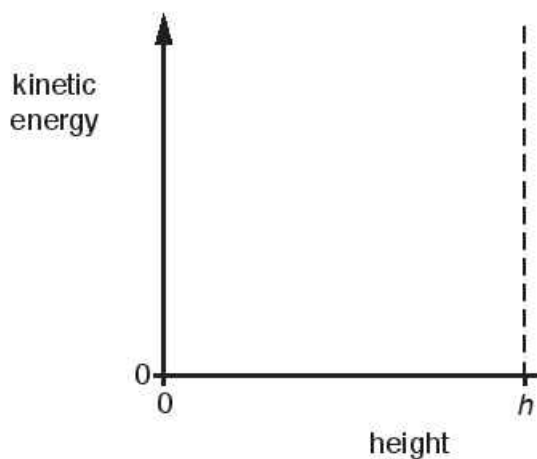


Fig. 2.2

[2]

- 31 The variation with extension x of the force F applied to a spring is shown in Fig. 4.1.

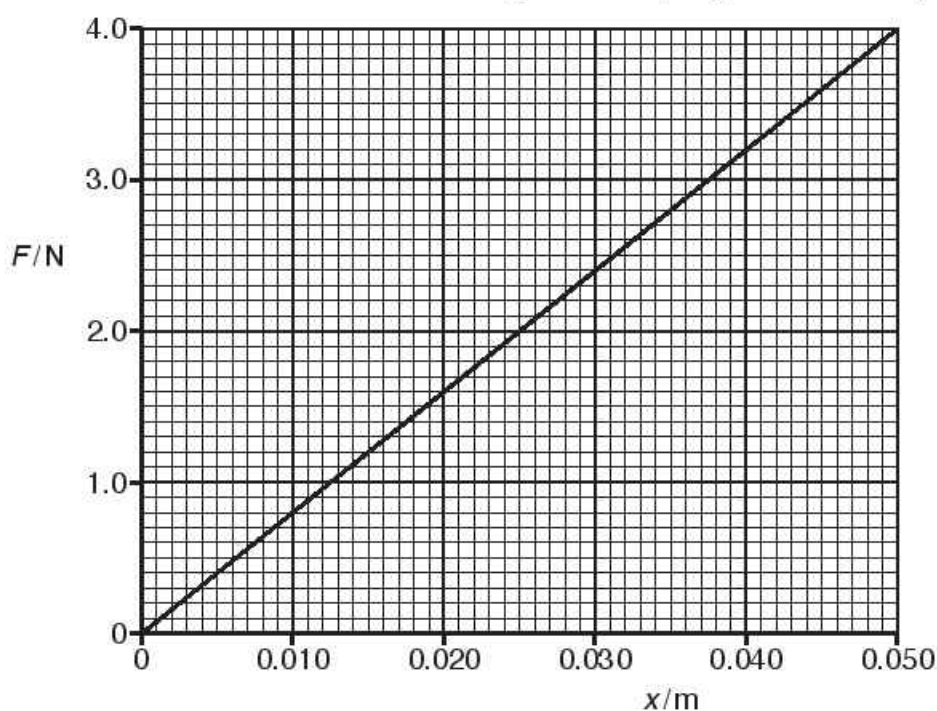


Fig. 4.1

The spring has an unstretched length of 0.080 m and is suspended vertically from a fixed point, as shown in Fig. 4.2.

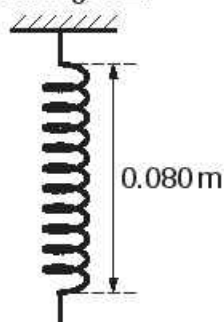


Fig. 4.2

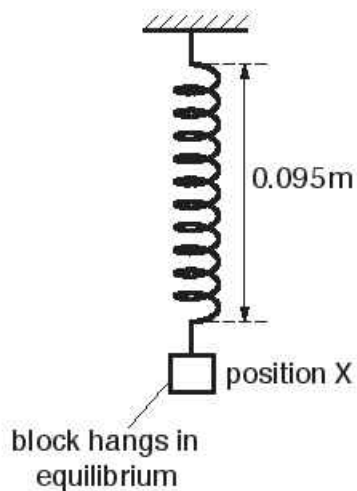


Fig. 4.3

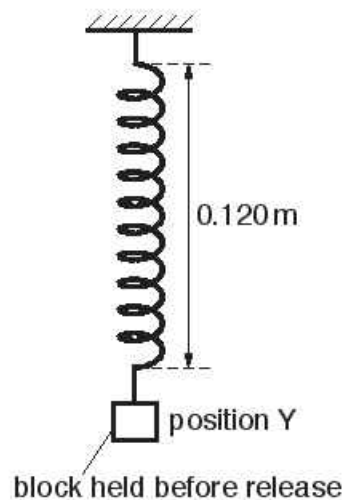


Fig. 4.4

A block is attached to the lower end of the spring. The block hangs in equilibrium at position X when the length of the spring is 0.095 m, as shown in Fig. 4.3.

The block is then pulled vertically downwards and held at position Y so that the length of the spring is 0.120 m, as shown in Fig. 4.4. The block is then released and moves vertically upwards from position Y back towards position X.

- (a) Use Fig. 4.1 to determine the spring constant of the spring.

spring constant = N m^{-1} [2]

- (b) Use Fig. 4.1 to show that the decrease in elastic potential energy of the spring is 0.055 J when the block moves from position Y to position X.

[2]

- (c) The block has a mass of 0.122 kg. Calculate the increase in gravitational potential energy of the block for its movement from position Y to position X.

increase in gravitational potential energy = J [2]

- (d) Use the decrease in elastic potential energy stated in (b) and your answer in (c) to determine, for the block, as it moves through position X:

- (i) its kinetic energy

kinetic energy = J [1]

- (ii) its speed.

speed = ms^{-1} [2]

- 32 (a) State Newton's third law of motion.

.....

.....

..... [2]

- (b) A block X of mass m_X slides in a straight line along a horizontal frictionless surface, as shown in Fig. 3.1.

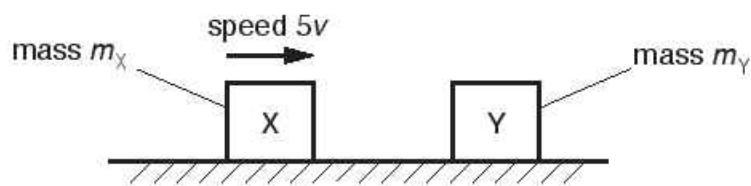


Fig. 3.1

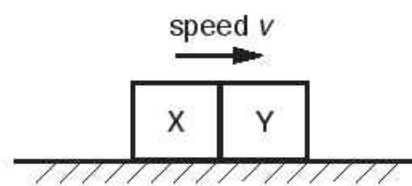


Fig. 3.2

The block X, moving with speed $5v$, collides head-on with a stationary block Y of mass m_Y . The two blocks stick together and then move with common speed v , as shown in Fig. 3.2.

- (i) Use conservation of momentum to show that the ratio $\frac{m_Y}{m_X}$ is equal to 4.

[2]

- (ii) Calculate the ratio

$$\frac{\text{total kinetic energy of X and Y after collision}}{\text{total kinetic energy of X and Y before collision}}.$$

ratio = [3]

(iii) State the value of the ratio in (ii) for a perfectly elastic collision.

ratio = [1]

(c) The variation with time t of the momentum of block X in (b) is shown in Fig. 3.3.

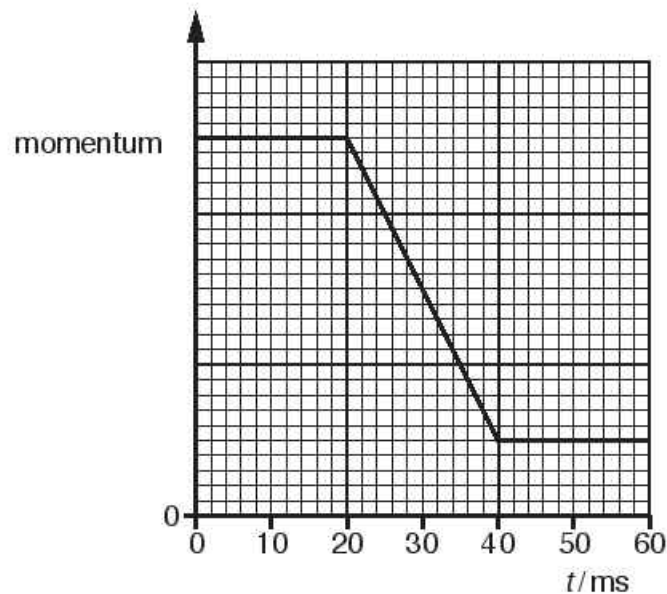


Fig. 3.3

Block X makes contact with block Y at time $t = 20$ ms.

(i) Describe, qualitatively, the magnitude and direction of the resultant force, if any, acting on block X in the time interval:

1. $t = 0$ to $t = 20$ ms

.....

2. $t = 20$ ms to $t = 40$ ms.

.....

.....

[3]

(ii) On Fig. 3.3, sketch the variation of the momentum of block Y with time t from $t = 0$ to $t = 60$ ms. [3]

33 (a) State **two** conditions for an object to be in equilibrium.

1.

.....

2.

.....

[2]

- (b) A sphere of weight 2.4 N is suspended by a wire from a fixed point P. A horizontal string is used to hold the sphere in equilibrium with the wire at an angle of 53° to the horizontal, as shown in Fig. 3.1.

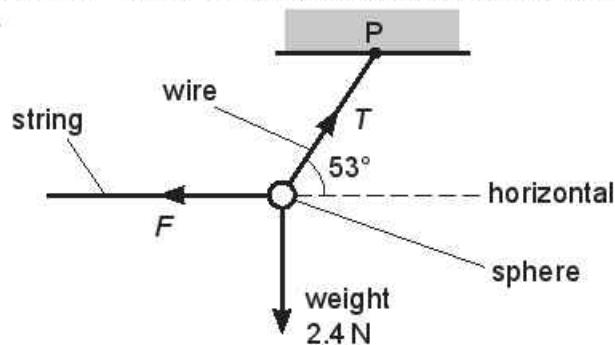


Fig. 3.1 (not to scale)

(i) Calculate:

1. the tension T in the wire

$$T = \dots\dots\dots \text{N}$$

2. the force F exerted by the string on the sphere.

$$F = \dots\dots\dots \text{N}$$

[2]

- (ii) The wire has a circular cross-section of diameter 0.50 mm. Determine the stress σ in the wire.

$$\sigma = \dots\dots\dots \text{Pa} \quad [3]$$

- (c) The string is disconnected from the sphere in (b). The sphere then swings from its initial rest position A, as illustrated in Fig. 3.2.

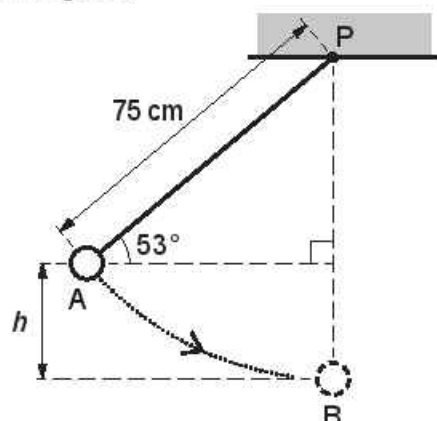


Fig. 3.2 (not to scale)

The sphere reaches maximum speed when it is at the bottom of the swing at position B. The distance between P and the centre of the sphere is 75 cm.

Air resistance is negligible and energy losses at P are negligible.

- (i) Show that the vertical distance h between A and B is 15 cm.

[1]

- (ii) Calculate the change in gravitational potential energy of the sphere as it moves from A to B.

change in gravitational potential energy = J [2]

- (iii) Use your answer in (c)(ii) to determine the speed of the sphere at B.
Show your working.

speed = m s^{-1} [3]

- 34 (a) Explain what is meant by *work done*.

.....
 [1]

- (b) A ball of mass 0.42 kg is dropped from the top of a building. The ball falls from rest through a vertical distance of 78 m to the ground. Air resistance is significant so that the ball reaches constant (terminal) velocity before hitting the ground. The ball hits the ground with a speed of 23 m s^{-1} .

- (i) Calculate, for the ball falling from the top of the building to the ground:

1. the decrease in gravitational potential energy

decrease in gravitational potential energy = J [2]

2. the increase in kinetic energy.

increase in kinetic energy = J [2]

- (ii) Use your answers in (b)(i) to determine the average resistive force acting on the ball as it falls from the top of the building to the ground.

average resistive force = N [2]

- (c) The ball in (b) is dropped at time $t = 0$ and hits the ground at time $t = T$. The acceleration of free fall is g .

On Fig. 3.1, sketch a line to show the variation of the acceleration a of the ball with time t from time $t = 0$ to $t = T$.

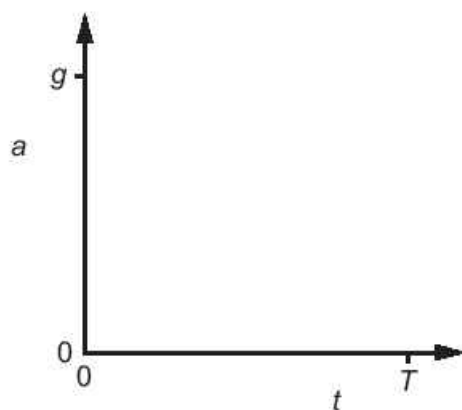


Fig. 3.1

[2]

35. 9702/22/F/M/21/Q2

- (a)** State what is meant by *work done*.

.....
..... [1]

- (b)** A beach ball is released from a balcony at the top of a tall building. The ball falls vertically from rest and reaches a constant (terminal) velocity. The gravitational potential energy of the ball decreases by 60 J as it falls from the balcony to the ground. The ball hits the ground with speed 16 m s^{-1} and kinetic energy 23 J.

- (i)** Show that the mass of the ball is 0.18 kg.

[2]

- (ii)** Calculate the height of the balcony above the ground.

height = m [2]

- (iii)** Determine the average resistive force acting on the ball as it falls from the balcony to the ground.

average resistive force = N [2]

- (c) State and explain the variation, if any, in the magnitude of the acceleration of the ball in (b) during the time interval when the ball is moving downwards **before** it reaches constant (terminal) velocity.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

36. 9702/21/M/J/21/Q2

A person uses a trolley to move suitcases at an airport. The total mass of the trolley and suitcases is 72 kg.

- (a) The person pushes the trolley and suitcases along a horizontal surface with a constant speed of 1.4 m s^{-1} and then releases the trolley. The released trolley moves in a straight line and comes to rest. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the power required to overcome the total resistive force on the trolley and suitcases when they move with a constant speed of 1.4 m s^{-1} .

power = W [2]

- (ii) Calculate the time taken for the trolley to come to rest after it is released.

time = s [3]

- (b) At another place in the airport, the trolley and suitcases are on a slope, as shown in Fig. 2.1.

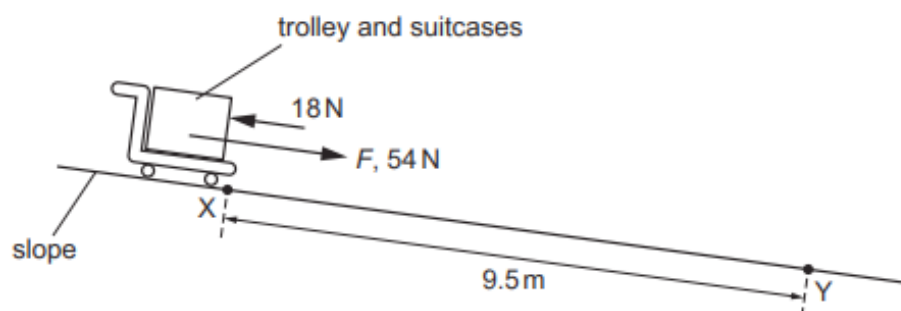


Fig. 2.1 (not to scale)

The person releases the trolley from rest at point X. The trolley moves down the slope in a straight line towards point Y. The distance along the slope between points X and Y is 9.5 m.

The component F of the weight of the trolley and suitcases that acts along the slope is 54 N. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the speed of the trolley at point Y.

speed = ms^{-1} [3]

- (ii) Calculate the work done by F for the movement of the trolley from X to Y.

work done = J [1]

- (iii) The trolley is released at point X at time $t = 0$.

On Fig. 2.2, sketch a graph to show the variation with time t of the work done by F for the movement of the trolley from X to Y.
Numerical values of the work done and t are not required.

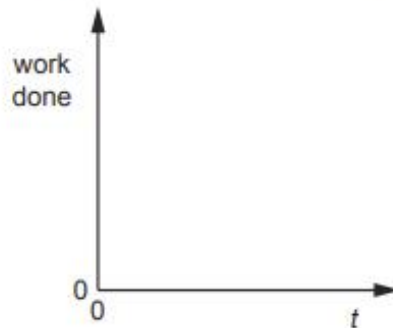


Fig. 2.2

[2]

- (c) The angle of the slope in (b) is constant. The frictional forces acting on the wheels of the moving trolley are also constant.

Explain why, in practice, it is incorrect to assume that the total resistive force opposing the motion of the trolley and suitcases is constant as the trolley moves between X and Y.

.....
..... [1]

[Total: 12]

37. 9702/22/M/J/21/Q3

A child of weight 330 N is at point X at the top of a slide. The slide is at the edge of a swimming pool, as shown in Fig. 3.1.

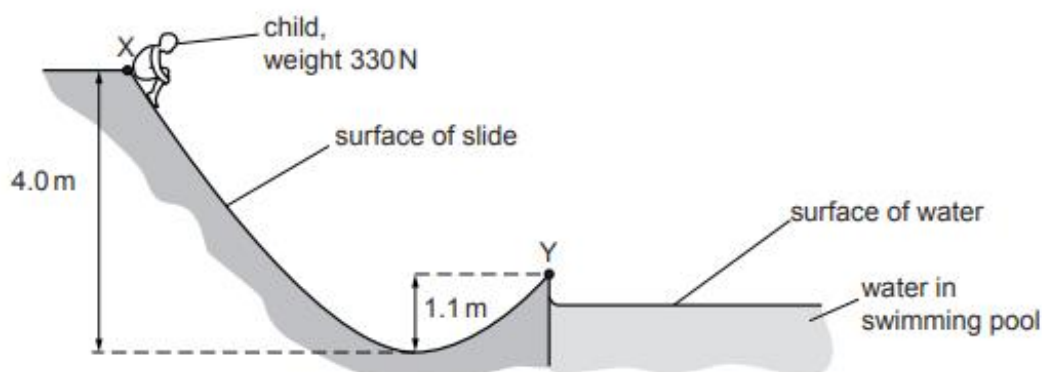


Fig. 3.1 (not to scale)

The child moves from rest to the lowest point of the slide that is a vertical distance of 4.0 m below X. The child continues moving towards point Y which is at the end of the slide and a vertical distance of 1.1 m above the lowest point. The kinetic energy of the child at Y is 540 J .

- (a) Calculate the difference in the gravitational potential energy of the child at points X and Y.

difference in gravitational potential energy = J [2]

- (b) An average frictional force of 52 N acts on the child when moving from X to Y.

By considering changes of energy, determine the distance moved by the child from X to Y.

distance moved = m [2]

- (c) The child leaves the slide at point Y with a velocity that is at an angle of 41° to the horizontal. The path of the child through the air is shown in Fig. 3.2.

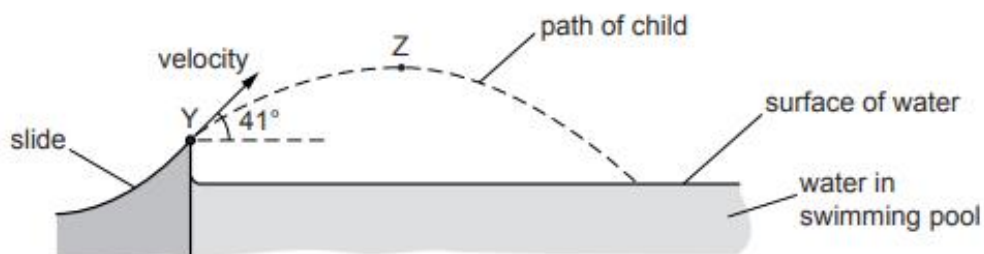


Fig. 3.2 (not to scale)

Point Z is the highest point on the path of the child through the air. Assume that air resistance is negligible.

Calculate the speed of the child at:

- (i) point Y

speed = ms^{-1} [2]

- (ii) point Z.

speed = ms^{-1} [2]

[Total: 8]

38. 9702/22/O/N/21/Q3

(a) Define *power*.

.....
 [1]

(b) A car of mass 1700 kg moves in a straight line along a slope that is at an angle θ to the horizontal, as shown in Fig. 3.1.

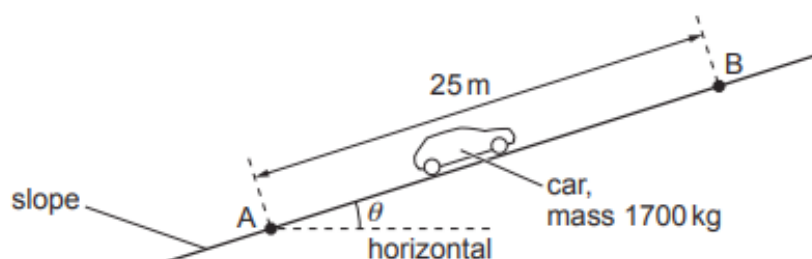


Fig. 3.1 (not to scale)

The car moves at constant velocity for a distance of 25 m from point A to point B.
 Air resistance and friction provide a total resistive force of 440 N that opposes the motion of the car.

For the movement of the car from A to B:

(i) state the change in the kinetic energy

change in kinetic energy = J [1]

(ii) calculate the work done against the total resistive force.

work done = J [1]

- (c) The movement of the car in (b) from A to B causes its gravitational potential energy to increase by $4.8 \times 10^4 \text{ J}$.

Calculate:

- (i) the increase in vertical height h of the car for its movement from A to B

$$h = \dots\dots\dots \text{ m [2]}$$

- (ii) angle θ .

$$\theta = \dots\dots\dots^\circ \text{ [1]}$$

- (d) The engine of the car in (b) produces an output power of $1.7 \times 10^4 \text{ W}$ to move the car along the slope.

Calculate the time taken for the car to move from A to B.

$$\text{time} = \dots\dots\dots \text{ s [2]}$$

[Total: 8]

39. 9702/22/F/M/22/Q4

A child moves down a long slide, as shown in Fig. 4.1.

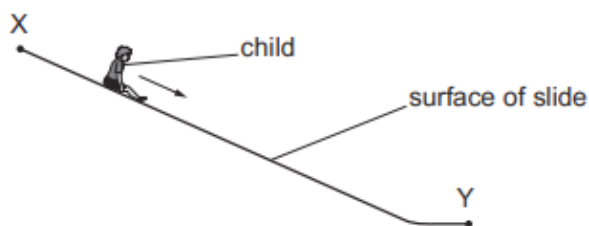


Fig. 4.1 (not to scale)

The child moves from rest at the top end X of the slide. An average resistive force of 76 N opposes the motion of the child as they move to the lower end Y of the slide. The kinetic energy of the child at Y is 300 J. The decrease in gravitational potential energy of the child as it moves from X to Y is 3200 J.

(a) Determine the ratio

$$\frac{\text{kinetic energy of the child at Y when the resistive force is 76 N}}{\text{kinetic energy of the child at Y if there is no resistive force}}.$$

ratio = [1]

(b) Use the answer in **(a)** to calculate the ratio

$$\frac{\text{speed of the child at Y when the resistive force is 76 N}}{\text{speed of the child at Y if there is no resistive force}}.$$

ratio = [2]

(c) Calculate the length of the slide from X to Y.

length = m [2]

- (d) At end Y of the slide, the child is brought to rest by a board, as shown in Fig. 4.2.

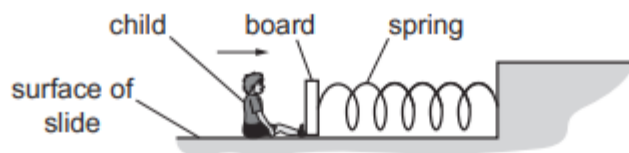


Fig. 4.2 (not to scale)

A spring connects the board to a fixed point. The spring obeys Hooke's law and has a spring constant of 63 N m^{-1} . The child hits the board so that it moves to the right and compresses the spring. The speed of the child becomes zero when the elastic potential energy of the spring has increased to its maximum value of 140 J .

- (i) Calculate the maximum compression of the spring.

maximum compression = m [2]

- (ii) Calculate the percentage efficiency of the transfer of the kinetic energy of the child to the elastic potential energy of the spring.

percentage efficiency = % [1]

- (iii) The maximum compression of the spring is x_0 . On Fig. 4.3, sketch a graph to show the variation of the elastic potential energy of the spring with its compression x from $x = 0$ to $x = x_0$. Numerical values are not required.

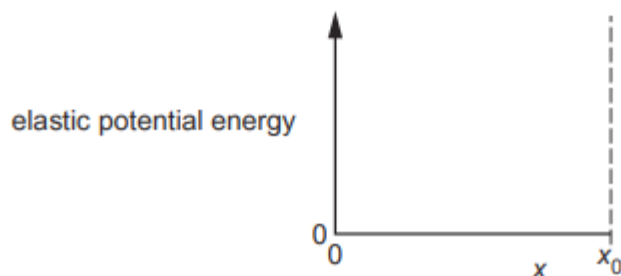


Fig. 4.3

[2]

40. 9702/21/M/J/22/Q3

(a) (i) Define power.

.....
 [1]

(ii) A force F takes time t to move an object through a displacement x at constant velocity v in the direction of the force. The work done by the force is W .

Use the definition of power to show that the power P transferred by the force is given by

$$P = Fv.$$

[2]

(b) A block is pulled up a slope by a wire attached to a motor, as shown in Fig. 3.1.

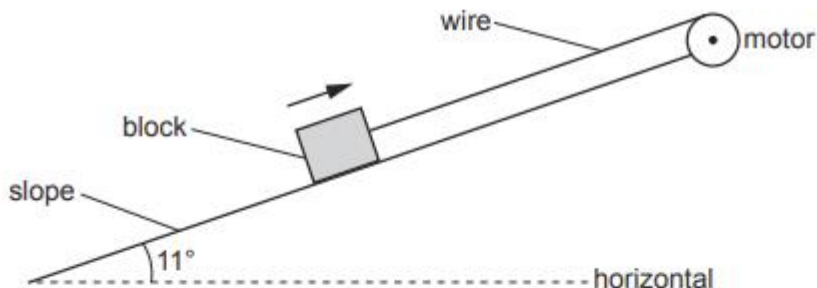


Fig. 3.1 (not to scale)

The useful power output of the motor is 56 W. The block has a weight of 430 N and travels with constant velocity along the slope at an angle of 11° to the horizontal.

Assume that there are no resistive forces opposing the motion of the block.

(i) Calculate the tension T in the wire.

$T =$ N [2]

- (ii) Calculate the speed of the block.

speed = ms^{-1} [1]

- (iii) The rate of increase of gravitational potential energy of the block is equal to the useful power output of the motor. One of the reasons for this is that there is no work done against resistive forces.

By considering the motion of the block, state another reason for this.

.....
..... [1]

- (iv) The motor has an efficiency of 80%.

Calculate the time taken for an input energy of 1.2 kJ to be supplied to the motor.

time taken = s [2]

[Total: 9]

41. 9702/23/M/J/22/Q4

- (a) A mass m moves a vertical distance Δh in a uniform gravitational field and gains gravitational potential energy ΔE_p . The acceleration of free fall is g .

Use the concept of work done to show that

$$\Delta E_p = mg\Delta h.$$

[2]

- (b) A 0.60 kg mass is attached to a string which is wrapped around the wheel of a generator, as shown in Fig. 4.1.

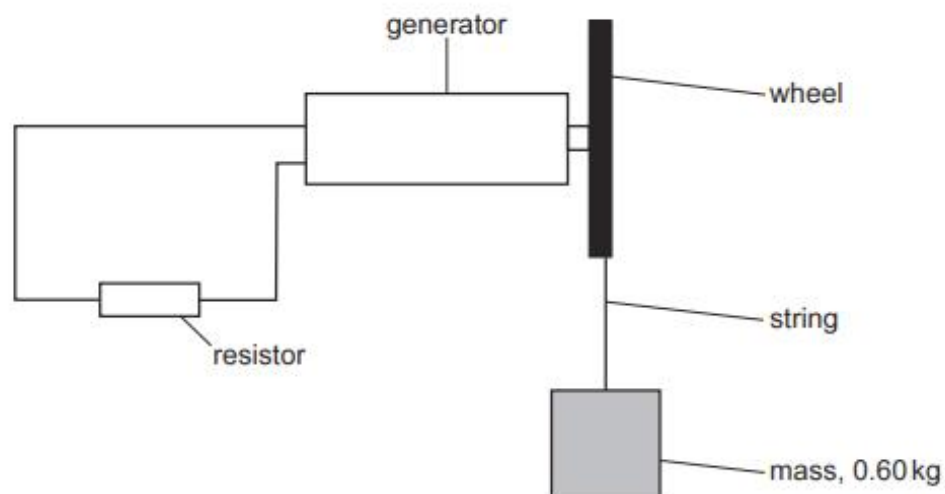


Fig. 4.1

The mass is held stationary above the floor. When released, the mass initially accelerates and then falls at a steady speed and spins the wheel. The generator causes a current in a resistor. Air resistance is negligible.

State the main energy change when the mass is falling at a steady speed.

..... energy to energy.

[1]

- (c) When falling at a steady speed, the mass in (b) falls through a vertical distance of 1.4 m in a time of 4.0 s. This causes a current of 90 mA in the resistor. The resistance of the resistor is $47\ \Omega$.

Calculate:

- (i) the rate of work done by the falling mass

rate of work done = W [2]

- (ii) the power dissipated in the resistor

power = W [2]

- (iii) the efficiency of the generator.

efficiency = [2]

[Total: 9]

42. 9702/21/O/N/22/Q3

(a) (i) Define power.

.....
..... [1]

(ii) Mechanical power P can be calculated using the formula $P = Fv$.

Use the concept of work and the definition of power to show how this formula is derived.

[2]

(b) The engine of a lorry provides 130 kW of power to the lorry's wheels when it is travelling at a constant speed of 25 m s^{-1} along a straight horizontal road.

Show that the resistive force opposing the forward motion of the lorry is 5200 N.

[11]

- (c) The lorry in (b) travels up a straight section of road that is inclined at an angle θ to the horizontal, as shown in Fig. 3.1.

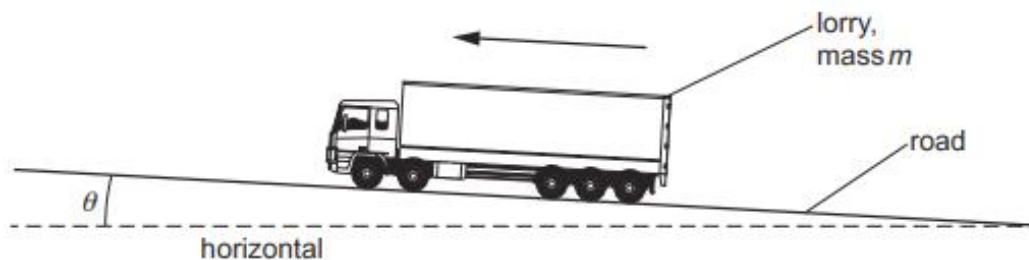


Fig. 3.1 (not to scale)

The lorry has mass m and the acceleration of free fall is g .

- (i) Determine an expression, in terms of m , g and θ , for the component of the weight of the lorry that acts parallel to the surface of the road.

[1]

- (ii) The total resistive force remains unchanged at 5200 N and the engine now provides greater power to maintain the speed of 25 m s^{-1} . The total mass m of the lorry is 36 000 kg. The angle θ is 1.4° .

Determine the power, in kW, now provided by the engine.

power = kW [3]

[Total: 8]

43. 9702/22/F/M/23/Q2

A motor uses a wire to raise a block, as illustrated in Fig. 2.1.

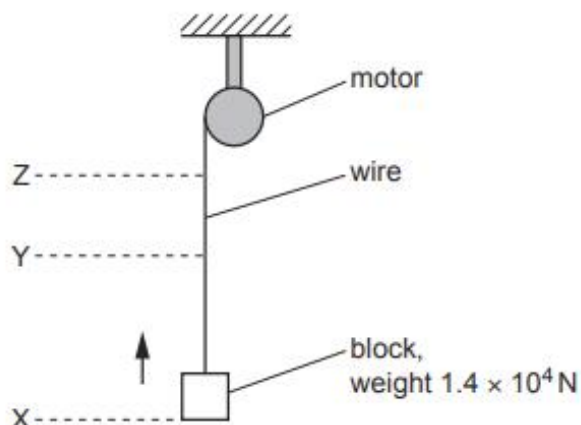


Fig. 2.1 (not to scale)

The base of the block takes a time of 0.49 s to move vertically upwards from level X to level Y at a constant speed of 0.64 m s^{-1} . During this time the wire has a strain of 0.0012 . The wire is made of metal of Young modulus $2.2 \times 10^{11} \text{ Pa}$ and has a uniform cross-section.

The block has a weight of $1.4 \times 10^4 \text{ N}$. Assume that the weight of the wire is negligible.

(a) Calculate:

(i) the cross-sectional area A of the wire

$$A = \dots\dots\dots \text{m}^2 \quad [2]$$

(ii) the increase in the gravitational potential energy of the block for the movement of its base from X to Y.

$$\text{increase in gravitational potential energy} = \dots\dots\dots \text{J} \quad [3]$$

- (b) The motor has an efficiency of 56%.

Calculate the input power to the motor as the base of the block moves from X to Y.

input power = W [3]

- (c) The base of the block now has a uniform deceleration of magnitude 1.3 m s^{-2} from level Y until the base of the block stops at level Z.

Calculate the tension T in the wire as the base of the block moves from Y to Z.

$T = \dots\dots\dots$ N [3]

- (d) The base of the block is at levels X, Y and Z at times t_X , t_Y and t_Z respectively.

On Fig. 2.2, sketch a graph to show the variation with time t of the distance d of the base of the block from level X. Numerical values of d and t are not required.

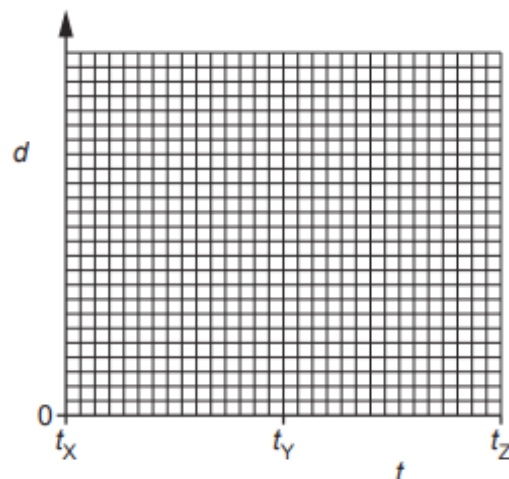


Fig. 2.2

[2]

[Total: 13]

44. 9702/22/M/J/23/Q2

- (a) State what is meant by the centre of gravity of an object.

.....
 [1]

- (b) Two blocks are on a horizontal beam that is pivoted at its centre of gravity, as shown in Fig. 2.1.

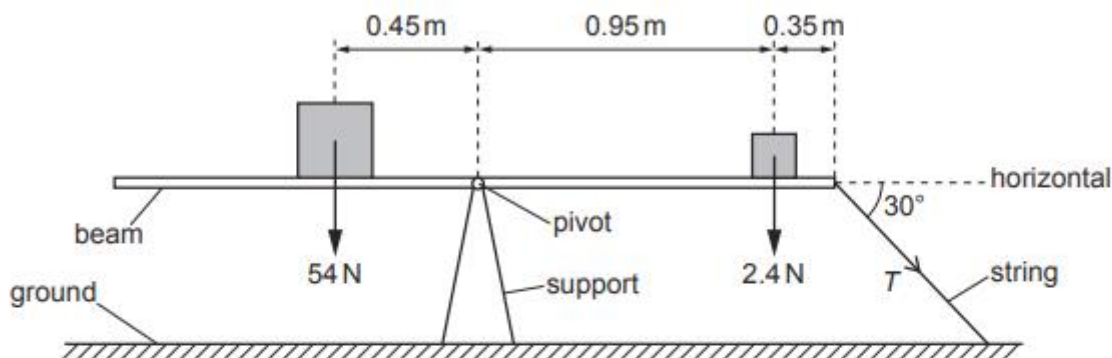


Fig. 2.1 (not to scale)

A large block of weight 54 N is a distance of 0.45 m from the pivot. A small block of weight 2.4 N is a distance of 0.95 m from the pivot and a distance of 0.35 m from the right-hand end of the beam.

The right-hand end of the beam is connected to the ground by a string that is at an angle of 30° to the horizontal. The beam is in equilibrium.

- (i) By taking moments about the pivot, calculate the tension T in the string.

$T = \dots\dots\dots$ N [3]

- (ii) The string is cut so that the beam is no longer in equilibrium.

Calculate the magnitude of the resultant moment about the pivot acting on the beam immediately after the string is cut.

resultant moment = $\dots\dots\dots$ Nm [1]

- (c) The beam in (b) rotates when the string is cut and the small block of weight 2.4 N is projected through the air. Fig. 2.2 shows the last part of the path of the block before it hits the ground at point Y.

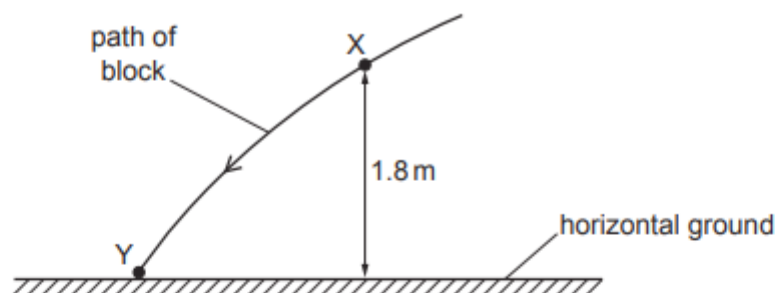


Fig. 2.2 (not to scale)

At point X on the path, the block has a speed of 3.4 ms^{-1} and is at a height of 1.8 m above the horizontal ground. Air resistance is negligible.

- (i) Calculate the decrease in the gravitational potential energy of the block for its movement from X to Y.

decrease in gravitational potential energy = J [2]

- (ii) Use your answer to (c)(i) and conservation of energy to determine the kinetic energy of the block at Y.

kinetic energy = J [3]

- (iii) State the variation, if any, in the direction of the acceleration of the block as it moves from X to Y.

..... [1]

- (iv) The block passes point X at time t_x and arrives at point Y at time t_y .

On Fig. 2.3, sketch a graph to show the variation of the magnitude of the horizontal component of the velocity of the block with time from t_x to t_y . Numerical values are not required.

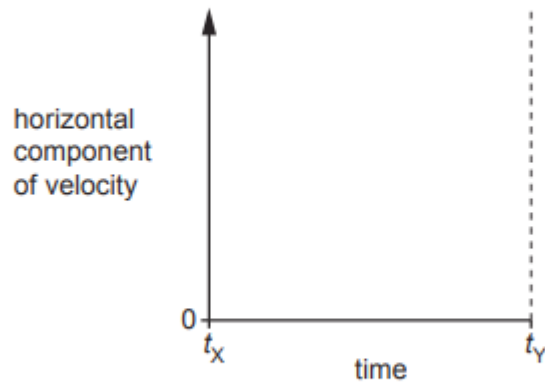


Fig. 2.3

[1]

[Total: 12]